PHYSICS

1) Where should be the 3rd charge + Q placed so that it remain in equilibrium :-



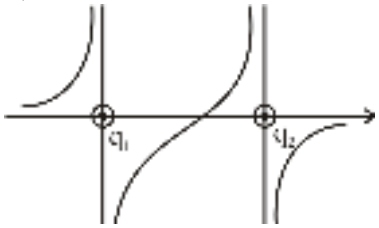
(1) x = 14 m

(2) x = 40 m

(3) x = 62 m

(4) x = 82 m

2) Electric field v/s distance graph is drawn then choose correct statement :-



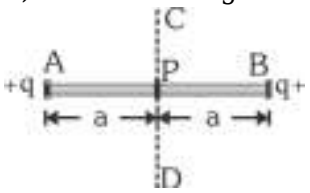
(1) q_1 and q_2 both are positive & $|q_1| > |q_2|$

(2) q_1 and q_2 both are negative & $|q_1| > |q_2|$

(3) q_1 is negative and q_2 is positive & $|q_1| > |q_2|$

(4) q_1 & q_2 both are negative and $|q_1| < |q_2|$

3) For the arrangement shown in the figure below, mark out the correct statement.



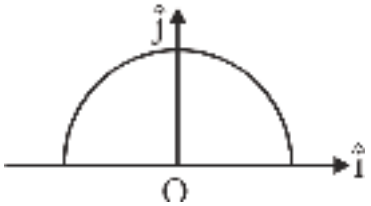
(1) For positive test charge, P is a position of stable equilibrium for displacement along AB.

(2) For positive test charge, P is a position of unstable equilibrium for displacement along CD.

(3) For negative test charge, P is a position of unstable equilibrium for displacement along AB.

(4) All of the above

4) A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net field $\vec{E}$ at the centre O is:-



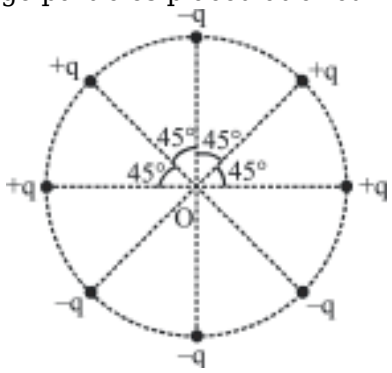
(1) $\frac{q}{4\pi^2\varepsilon_0 r^2}\hat{j}$

(2) $-\frac{q}{4\pi^2\varepsilon_0 r^2}\hat{j}$

(3) $-\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$

(4) $\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$

5) Eight charge particles placed at circumference of circle of radius r. The net electric field at centre



of circle is :-

(1) zero

(2) $\frac{2\sqrt{2}kq}{r^2}$

(3) $\frac{2kq}{r^2}$

(4) $\frac{\sqrt{2}kq}{r^2}$

6) How many electrons must be removed from a silver coin so that a 2.4 C charge remains on it ?

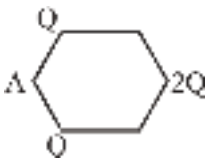
(1) 2.5×10^{19}

(2) 1.5×10^{19}

(3) 1.5×10^{-19}

(4) 2.5×10^{-19}

7) Regular hexagon of side 'a' is shown in fig. find electric field intensity at point A :-



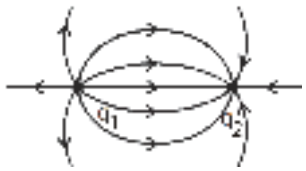
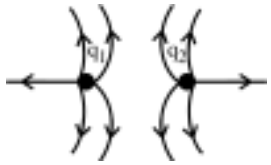
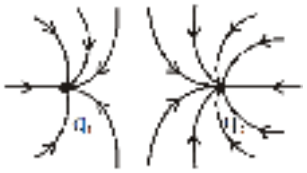
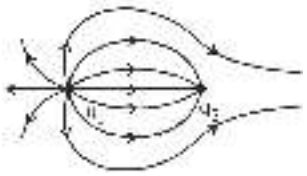
(1) $\frac{KQ}{a^2}$

(2) $\frac{3}{2}\frac{KQ}{a^2}$

(3) $\frac{2KQ}{a^2}$

(4) $\frac{5}{4}\frac{KQ}{a^2}$

8) List–I shows the electric field lines diagram for two point charges q_1 (left) and q_2 (right). List–II has the possible information about the charges q_1 and q_2. Match the correct combination of charges with their field line diagram.

List-I		List-II	
(P)		(1)	$\|q_1\| > \|q_2\|$
(Q)		(2)	$q_1 > 0$, $q_2 < 0$
(R)		(3)	$q_1 > 0$, $q_2 > 0$
(S)		(4)	$q_1 < 0$, $q_2 < 0$
		(5)	$\|q_1\| < \|q_2\|$

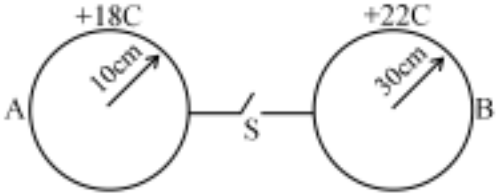
(1) (P) - 3 ; (Q) - 3 ; (R) - 3,5 ; (S) - 4,5

(2) (P) - 2 ; (Q) - 3 ; (R) - 1,2 ; (S) - 4,5

(3) (P) - 2 ; (Q) - 3 ; (R) - 4,5 ; (S) - 1,2

(4) (P) - 3 ; (Q) - 4 ; (R) - 1,2 ; (S) - 3,5

9) On closing the switch, charge flowing through the wire will be –



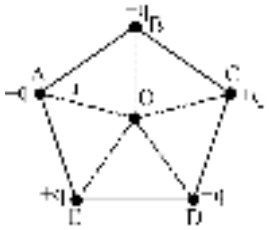
(1) + 8C from A to B

(2) + 10C from B to A

(3) – 7C from B to A

(4) – 15C from B to A

10) Five point charges are placed at corners of a regular pentagon as shown in figure. Net force on

an electron placed at centre of pentagon will be – 

(1) $\frac{kqe}{r^2}$ along $\overrightarrow{OA}$

(2) $\frac{kqe}{r^2}$ along $\overrightarrow{AO}$

(3) $\frac{2kqe}{r^2}$ along $\overrightarrow{OA}$

(4) $\frac{2kqe}{r^2}$ along $\overrightarrow{AO}$

11) A drop of water 10^{-6} kg of carries 10^{-6} C charge. What electric field should be applied to balance its weight (assume g = 10m/s^2)

(1) 10 V/m upward

(2) 10 V/m downward

(3) 0.1 V/m downward

(4) 0.1 V/m upward

12) The specific charge on an electron is :-

(1) 1.8×10^{-19} C/kg

(2) 1.8×10^{-12} C/kg

(3) 1.8×10^{11} C/kg

(4) More than one of the above

13) A uniformly charged ring of radius R has charge Q. Electric field at axis of ring at distance $\sqrt{3}R$ from its centre is :-

(1) $\sqrt{3}\frac{kQ}{R^2}$

(2) $\frac{2}{3\sqrt{3}}\frac{kQ}{R^2}$

(3) $\frac{\sqrt{3}}{8}\frac{kQ}{R^2}$

(4) $\frac{\sqrt{3}}{16}\frac{kQ}{R^2}$

14)

Number of electrons in one coulomb of charge will be :-

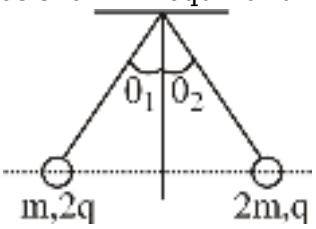
(1) 5.46×10^{29}

(2) 6.25×10^{18}

(3) $1.6 \times 10^{+19}$

(4) 9×10^{11}

15) Two particles which have masses and charges (m, 2q) and (2m, q) respectively are hang by massless threads from a point as shown in equilibrium. Particles are on same horizontal line. What can be the relation in θ_1 & θ_2 ?



(1) $\theta_1 > \theta_2$

(2) $\theta_1 < \theta_2$

(3) $\theta_1 = \theta_2$

(4) More information is required.

16) Potential energy of a particle moving along x-axis is given as $U = (x^3 + 3x^2 - 9x + 6)$ J. Then find minimum possible value of U

(1) –21J

(2) –4J

(3) 1J

(4) –3J

17) Find $\left(1-\frac{1}{3}+\frac{1}{9}-\frac{1}{27}+.....\infty\right)$:-

(1) $\frac{1}{3}$

(2) $\frac{2}{3}$

(3) $\frac{3}{2}$

(4) $\frac{3}{4}$

18) A particle moves along the straight line $2y = 3x + 5$. Which coordinate changes at a faster rate ?

(1) x–coordinate

(2) y–coordinate

(3) Both x and y coordinates

(4) Data insufficient

19) The approximate value of x where
x = (sin 2°) (cos 2°), is :

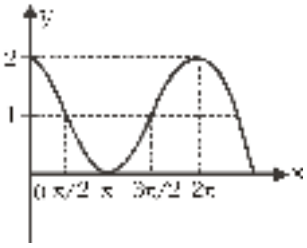
(1) $\frac{\pi}{90}$

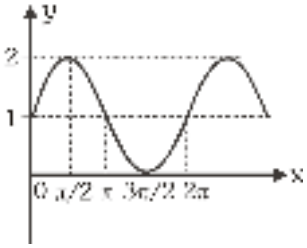
(2) 2

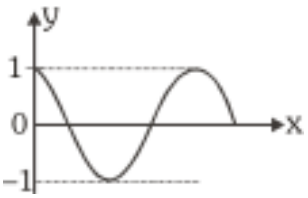
(3) 1

(4) $\frac{\pi}{45}$

20) The graph of function y = 1 + cos x will be

(1) 

(2) 

(3) 

(4) None of these

21) Five forces $\vec{F}_1$, $\vec{F}_2$, $\vec{F}_3$, $\vec{F}_4$ and $\vec{F}_5$ acting on a particle of mass 2.0 kg so that, it is moving with 4 m/s^2 in east direction. If $\vec{F}_1$ force is removed, then the acceleration becomes 7 m/s^2 in north, then the acceleration of the block if only $\vec{F}_1$ is acting will be :

(1) 16 m/s^2

(2) $\sqrt{65}$ m/s^2

(3) $\sqrt{260}$ m/s^2

(4) $\sqrt{33}$ m/s^2

22) Resultant of two vectors $\vec{A}$ and $\vec{B}$ is perpendicular to A and magnitude of A is half of B then angle between 'A' and 'B' is :

(1) 120°

(2) 150°

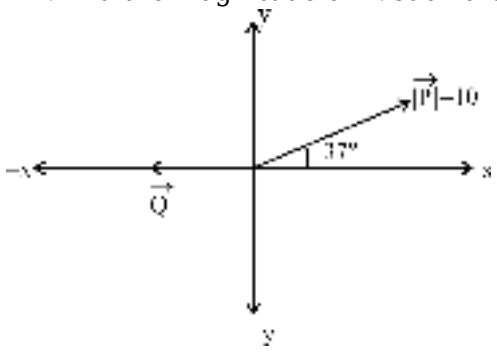
(3) 135°

(4) 115°

23) If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors such that $\vec{a}+\vec{b}-\vec{c}=0$, then the angle between $\vec{a}$ and $\vec{b}$ is

(1) 30°
(2) 60°
(3) 90°
(4) 120°

24) For the given $\vec{P}$ & $\vec{Q}$ find the magnitude of $\vec{Q}$ such that resultant is along y-axis :-



(1) 6
(2) 10
(3) 8
(4) 5

25) What is the angle between $\hat{i}+\hat{j}+\hat{k}$ and $\hat{i}$?

(1) $\frac{\pi}{6}$
(2) $\frac{\pi}{4}$
(3) $\frac{\pi}{3}$
(4) None of these

26) If $\vec{r}=3\hat{i}+4\hat{j}-\hat{k}$, $\vec{F}=\hat{i}+\hat{j}-\hat{k}$ then torque ($\vec{\tau}$) will be : $(\vec{\tau}=\vec{r}\times\vec{F})$

(1) $3\hat{i}+2\hat{j}+\hat{k}$
(2) $-3\hat{i}+2\hat{j}+\hat{k}$
(3) $-3\hat{i}+2\hat{j}-\hat{k}$
(4) $-3\hat{i}-2\hat{j}+\hat{k}$

27) A particle is projected at an angle θ above the horizontal with a speed u. After some time the direction of its velocity makes an angle ϕ above the horizontal. The speed of the particle at this instant is :-

(1) $u\cos\theta$

(2) $\frac{u\cos\theta}{\cos\phi}$

(3) $\frac{u\sin\theta}{\sin\phi}$

(4) $u\cos\phi$

28) Ratio of minimum kinetic energies of two projectiles of same mass is 4 : 1. The ratio of the maximum height attained by them is also 4 : 1. The ratio of their ranges would be :-

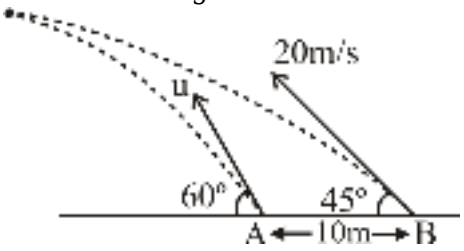
(1) 16 : 1

(2) 4 : 1

(3) 8 : 1

(4) 2 : 1

29) Two projectiles are projected at same time as shown in fig. What should be the velocity 'u' of projectile A so that they can collide in the air:-



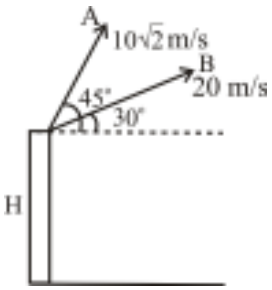
(1) $\frac{20}{\sqrt{2}}$

(2) $20\sqrt{\frac{2}{3}}$

(3) $\frac{20}{\sqrt{3}}$

(4) $20\sqrt{3}$

30) Two balls are thrown from a tower as shown in the figure :-



Which of the ball will hit the ground first :-

(1) A

(2) B

(3) Both will strike at the same time

(4) Depends on height of tower

31) Two particles A and B are located in x-y plane at points (0, 0) and (0, 4m). They simultaneously

start moving with velocities $\vec{v}_A = 2\hat{j}$ m/s and $\vec{v}_B = 2\hat{i}$ m/s :

(1) At t = 2s the minimum distance between them is $2\sqrt{2}$ m

(2) At t = 2s the minimum distance between them is $4\sqrt{2}$ m

(3) At t = 1s the minimum distance between them is $2\sqrt{2}$ m

(4) At t = 1s the minimum distance between them is $4\sqrt{2}$ m

32) Rain drops are falling with velocities $(2\hat{i} - 4\hat{j})$ m/sec what should be the velocities of a man so that rain drops hit him with speed 5m/sec.

(1) $-\hat{i}$

(2) $5\hat{i}$

(3) $2\hat{i}$

(4) Both (1) and (2)

33) A particle moves in x-y plane according to equations x = αt and y = αt(1–βt), where α and β are positive constants and t is time. Then equation of the trajectory(path) of the partical is :-

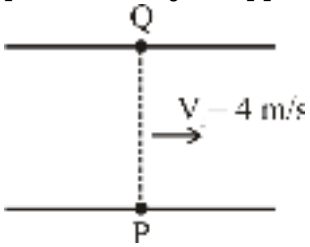
(1) $y = x\left(1 - \frac{\alpha}{\beta}x\right)$

(2) y = αx(1–βx)

(3) $y = x\left(1 - \frac{\beta}{\alpha}x\right)$

(4) y = βx(1–αx)

34) A boat man could row his boat with speed 10 m/s in still water. He want to take his boat from point P to Q just opposite on other bank. If river flowing at speed 4 m/s then he sould row his boat :-



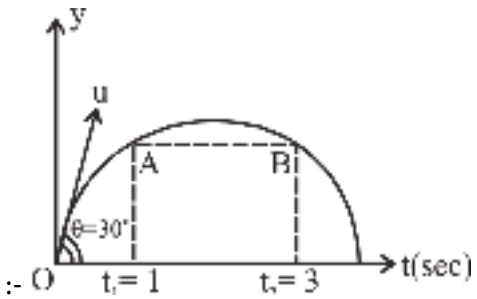
(1) at right angle to stream

(2) at an angle $\sin^{-1}\left(\frac{2}{5}\right)$ with PQ up the stream

(3) at an angle $\sin^{-1}\left(\frac{2}{5}\right)$ with PQ down the stream

(4) at an angle $\cos^{-1}\left(\frac{2}{5}\right)$ with PQ down the stream

35) A particle is thrown with velocity u m/sec. At an angle of projection 30° from horizontal, It passes A and B as shown in figure at time t_1 = 1 sec and t_2 = 3 sec. The value of u is (g = 10 m/sec^2)



(1) 20 m/sec

(2) 10 m/sec

(3) 40 m/sec

(4) 5 m/sec

36) There are two bodies A & B of same mass. Body A is at rest while body B is under uniform motion. Which of the following is correct statement ?

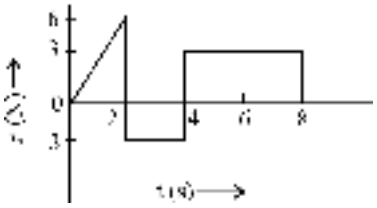
(1) Inertia of A > inertia of B

(2) Inertia of A < inertia of B

(3) Inertia of A = inertia of B

(4) Either 1st, 2nd or 3rd depending upon the shape of body

37) The force 'F' acting on a particle of mass 'm' is indicated by the force-time graph shown below. The change in momentum of the particle over the time interval from zero to 8 s is :-



(1) 24 Ns

(2) 20 Ns

(3) 12 Ns

(4) 6 Ns

38) Bullets of mass 50g are fired from a gun of mass 10 kg with velocity of 300 m/s. If 5 bullets are fired per second then what force a person will have to apply on the gun so that it does not recoil:-

(1) 75 N

(2) 150 N

(3) 225 N

(4) 300 N

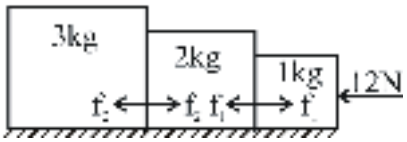
39) A machine gun fires a bullet of mass 40 g with a velocity 1200 m/s. The man holding it, can exert maximum force of 144 N on the gun. How many bullets can he fire per second at the most ?

(1) 1

(2) 2

(3) 3

(4) 4

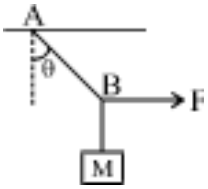
40) Contact forces f_1 and f_2 are :- 

(1) 10N, 6N

(2) 6N, 10N

(3) 3N, 2N

(4) 2N, 3N

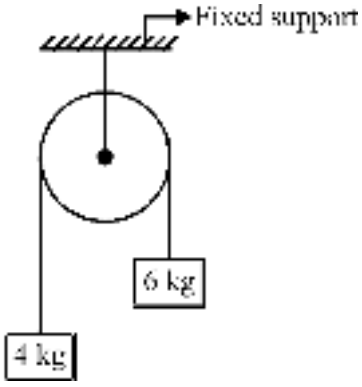
41) If the system is in equilibrium then tension in the string AB is 

(1) F sinθ

(2) F/sinθ

(3) F cos θ

(4) $\frac{F}{\cos\theta}$

42) For the given pulley system with ideal pulley and ideal strings, find the acceleration of the 4 kg block and tension in string connecting the blocks. (g = 10 m/s^2) :- 

(1) 2 m/s^2, 32 N

(2) 2 m/s^2, 48 N

(3) 4 m/s^2, 32 N

(4) 4 m/s^2, 48 N

43) A light string passing over a smooth light pulley connects two blocks of masses m_1 and m_2 (where $m_2 > m_1$). If the acceleration of the system is $\frac{g}{\sqrt{2}}$, then the ratio of the masses $\frac{m_1}{m_2}$ is :

(1) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$

(2) $\frac{1+\sqrt{5}}{\sqrt{5}-1}$

(3) $\frac{1+\sqrt{5}}{\sqrt{2}-1}$

(4) $\frac{\sqrt{3}+1}{\sqrt{2}-1}$

44) A body of weight 200 N is suspended from a tree branch though a chain of mass 10 kg. The branch pulls the chain by a force equal to (if g = 10 m/s^2):

(1) 150 N

(2) 300 N

(3) 200 N

(4) 100 N

45) A player caught a cricket ball of mass 150 g moving at a speed of 20 m/s. If the catching process is completed in 0.1 s, the magnitude of force exerted by the ball on the hand of the player is:

(1) 150 N

(2) 3 N

(3) 30 N

(4) 300 N

CHEMISTRY

1) Which of the following molecules have unequal bond lengths ?

(1) SF_4

(2) SiF_4

(3) SF_6

(4) NF_3

2) Which of the following is not correct about a co-ordinate bond :-

(1) A co-ordinate bond once formed cannot be distinguished from a covalent bond

(2) A co-ordinate bond is also called a semi polar bond

(3) A co-ordinate bond is non-directional in nature

(4) Due to co-ordinate bond the formal charges on donar and acceptor atoms are + and – respectively.

3) How many polar molecules are present in the following molecules ?
ClF_3, BrF_5, H_2O, XeF_2, CO_2, $SnCl_4$, XeF_6, NH_3, NF_3, PCl_2F_3

(1) 5

(2) 6

(3) 7

(4) 8

4) Correct order of boiling point is :-

(1) $I_2 > Cl_2 > Br_2$

(2) $He > H_2$

(3) $ICl > Br_2$

(4) $H_2 > D_2 > T_2$

5) Pick out the incorrect match of intermolecular attraction between molecule/ion in the following pairs

(1) HBr and H_2S–Dipole-dipole attraction

(2) Cl_2 and CCl_4–Dispersion force (london force)

(3) NH_3 and C_6H_6–Hydrogen bond

(4) I_2 and NO_3^-–Ion-induced dipole attraction

6) O_2F_2 is an unstable yellow orange solid and H_2O_2 is a colourless liquid, both haveO–O bond. O–O bond length in H_2O_2 & O_2F_2 is respectively.

(1) 1.22Å, 1.48Å

(2) 1.48Å, 1.22Å

(3) 1.22Å, 1.22Å

(4) 1.48Å, 1.48Å

7)

In nitrate ion type of π bond is :

(1) pπ-dπ

(2) dπ-dπ

(3) Both (1) and (2)

(4) pπ-pπ

8) Select the incorrect order of Bond order :-

(1) $O_2^+ > O_2$

(2) $B_2^- > B_2$

(3) $C_2^- > C_2$

(4) $N_2^+ > N_2$

9)

Which σ bond has highest bond energy :

(1) F–F

(2) O–O

(3) N–N

(4) C–C

10) **Assertion :** H_2^+ is paramagnetic in nature.
Reason : Bond order of H_2^+ is 0.5.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are false.

11) **Assertion :** Boiling point of HF is lesser than H_2O.
Reason : Extent of hydrogen bonding in HF is lesser than H_2O.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

12) Match the following (where $\theta \Rightarrow$ bond angle) :-

	Column-I		**Column-II**
(A)	$100° < \theta < 109°28'$	(p)	H_2O
(B)	$90° < \theta < 100°$	(q)	SO_3
(C)	$\theta > 109°\ 28'$	(r)	PH_3
(D)	$\theta > 120°$	(s)	NO_2

(1) (A)→r, (B)→p, (C)→s, (D)→q

(2) (A)→p, (B)→s, (C)→q, (D)→r

(3) (A)→p, (B)→s, (C)→r, (D)→q

(4) (A)→p, (B)→r, (C)→q, (D)→s

13) How many statements given are true
(i) bond angle of NH_3 is greater than AsH_3
(ii) Dipole moment of CH_3F is greater than CH_3Cl
(iii) Intramolecular H-bonding decrease water solubility.
(iv) Geometrical shape of NH_3 & BF_3 are same
(v) Diatomic boron is paramagnetic

(1) i, ii, v

(2) ii, iv, v

(3) i, iii, v

(4) i, ii, iii, v

14) In which bond angle is maximum

(1) NO_2^-
(2) NO_2
(3) NO_3^-
(4) All equal

15) Correct stability order of N_2 and its ion will be-

(1) $N_2 > N_2^+ > N_2^- > N_2^{-2}$
(2) $N_2^- > N_2^+ > N_2 > N_2^{-2}$
(3) $N_2^+ > N_2^- > N_2 > N_2^{-2}$
(4) $N_2 > N_2^+ = N_2^- > N_2^{-2}$

16) In which pair two species are not isostructural-

(1) SiF_4 and SF_4
(2) IO_3^- and XeO_3
(3) PF_6^- and SF_6
(4) BH_4^- and NH_4^+

17) Which order of boiling point is correct-

(1) $SbH_3 > NH_3 > PH_3 > AsH_3$
(2) $NH_3 > SbH_3 > AsH_3 > PH_3$
(3) $SbH_3 > NH_3 > AsH_3 > PH_3$
(4) $AsH_3 > NH_3 > SbH_3 > PH_3$

18) In $C_2(CN)_4$ C-atoms are-

(1) sp hybridised
(2) sp^2 hybridised
(3) sp and sp^3 hybridised
(4) sp & sp^2 hybridised

19) Which compound is polar but have almost nonpolar bond :

(1) CS_2
(2) PH_3
(3) NH_3
(4) H_2O

20) No. of anti boding electrons in O_2^{-2} will be-

(1) 8
(2) 6
(3) 4
(4) 2

21) In PO_4^{-3} ion P–O bond order and lone pair on P will be respectively-

(1) 1.75, 0
(2) 1.25, 0
(3) 1.25, 1
(4) 1.75, 1

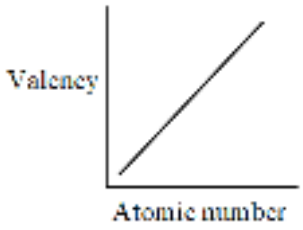
22) Which of the following is planar compound-

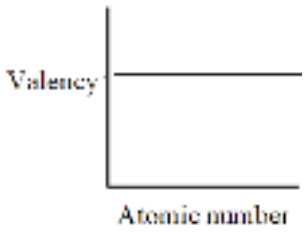
(1) XeO_4
(2) XeF_6
(3) XeO_2F_2
(4) XeF_4

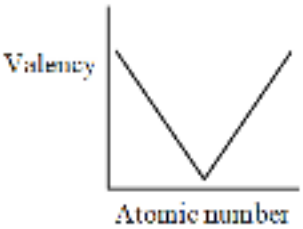
23) **Assertion:** In ClF_3 90° bond angle are present
Reason: ClF_3 is perfect T-shape molecule.

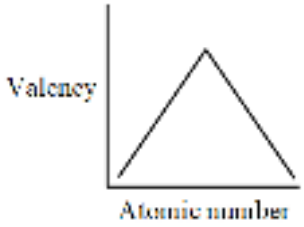
(1) A true R false
(2) A false R true
(3) Both A & R are false
(4) Both A & R are true

24) Which graph represents trend of valency with respect to hydrogen in a period-

(1)


(2)


(3)


(4) 

25) Which statement is incorrect.

(1) $N(CH_3)_3$ is more basic than $N(SiH_3)_3$
(2) p-nitrophenol is more volatile than o-nitrophenol
(3) Hydronium ion in water shows ion dipole attraction
(4) In KHF_2 symmetrical H-bond is present.

26) Example of complete octet is-

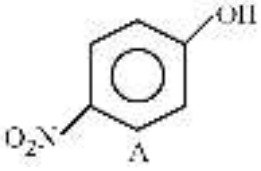
(1) $BeCl_2$ solid-polymer
(2) BeH_2 solid-polymer
(3) $AlCl_3$-monomer
(4) $BeCl_2$ solid-dimer

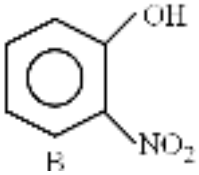
27) Among NO_3^-, N_2O_4, NO_2, HF_2^-, BF_3, SiO_2, CO_2
how many species shows resonance -

(1) 5
(2) 4
(3) 7
(4) 6

28) During change of C_2 to C_2^- ion, the electron adds on which one of the following orbitals ?

(1) σ*
(2) σ
(3) π*
(4) π

29) Between the two compounds shown below, the vapour pressure of B at a particular temperature is expected to be : 

A and 

B

(1) Higher than that of A
(2) Lower than that of A
(3) Same as that of A
(4) Can be higher or lower depending upon that size of the vessel

30) What is volume (in L) of oxygen required at STP to completely convert 1.5 moles of sulphur to sulphur dioxide ?
$S + O_2 \rightarrow SO_2$

(1) 43.6

(2) 33.6

(3) 11.2

(4) 23.6

31) Equal masses of O_2 and N_2 are reacted according to the equation : $O_2 + N_2 \rightarrow 2NO$. Which statement is true ?

(1) O_2 is the limiting reagent and N_2 is present in excess

(2) N_2 is the limiting reagent and O_2 is present in excess.

(3) All of O_2 and N_2 react and neither is in excess.

(4) Nothing can be said about the limiting reagent

32) How many grams of $AgNO_3$ are needed to prepare 500 mL of a 0.240 M solution ? ($AgNO_3$ = 170 g/mol)

(1) 1.42 g

(2) 20.4 g

(3) 120 g

(4) 354 g

33)

What is the change in volume on complete decomposition of 10 L NH_3 gas into $N_2(g)$ and $H_2(g)$ (Assume temperature and pressure are constant) ?

(1) – 20 L

(2) + 20 L

(3) + 10 L

(4) – 10 L

34) An organic compound containing C,H,N and O weighing 7.3 g have 3.6 g carbon, 0.7 g hydrogen, 1.4 g nitrogen and rest oxygen. Write the empirical formula of the compound :-

(1) C_3H_7NO

(2) $C_4H_8N_2O$

(3) $C_6H_{14}N_2O_2$

(4) $C_3H_7N_2O$

35) What mass of $CaCO_3$ sample is needed which is 80% pure to produce 1.12L of CO_2 at STP if yield of reaction is 50% :-

(1) 10 g

(2) 3.175 g

(3) 12.5 g

(4) 20 g

36) In a compound A containing carbon, hydrogen and oxygen, the % of hydrogen is one sixth that of carbon which is 1.5 times that of oxygen. What is the simplest formula for the compound:-

(1) $C_2H_4O_2$

(2) C_2H_4O

(3) C_2H_6O

(4) CH_4O

37) Two oxides of a metal contain 22.22% and 30% oxygen by mass respectively. If the formula of the first oxide is MO, then the formula of the second oxide is :-

(1) MO

(2) MO_2

(3) M_2O_3

(4) M_2O

38) The formula of an hypothetical gas is found to be $(N_2O)_x$. If vapour density of gas is 220, then find value of x :-

(1) 5

(2) 10

(3) 2.0

(4) 15

39) At room temperature and pressure, two flask of equal volume are filled with H_2 and SO_2 separately. Particles which are equal in number in the flasks are :-

(1) Atoms

(2) Electrons

(3) Molecules

(4) Neutrons

40) A certain compound has the molecular formula X_4O_6. If 10 g of X_4O_6 has 5.72 g of X, atomic mass of X is :

(1) 32 amu

(2) 37 amu

(3) 42 amu

(4) 98 amu

41) 40 milligram diatomic volatile substance (X_2) is converted to vapour that displaced 4.92 mL of air at 1 atm and 300 K. Atomic weight of element X is nearly :-

(1) 400

(2) 240

(3) 200

(4) 100

42)

Raoult’s law is obeyed by each constituent of a binary liquid solution when :

(1) the forces of attractions between like molecules are greater than those between unlike molecules.

(2) the forces of attractions between like molecules are smaller than those between unlike molecules.

(3) the forces of attractions between like molecules are identical with those between unlike molecules.

(4) the volume occupied by unlike molecules are different.

43) 5 mL of acetone is mixed with 100 mL of H_2O. The vapour pressure of water above the solution is :-

(1) Equal to the vapour pressure of pure water

(2) Equal to the vapour pressure of the solution

(3) Less than the vapour pressure of pure water

(4) More than the vapour pressure of pure water

44) Which one is incorrect for ideal solutions

(1) $\Delta S_{mix.} = 0$

(2) $\Delta G_{mix.} = 0$

(3) They forms azeotropes

(4) All are incorrect

45) An azeotropic solution of two liquids has a boiling point lower than either of them when it :-

(1) Shows negative deviation from Raoult's law

(2) Shows no deviation from Raoult's law

(3) Shows positive deviation from Raoult's law

(4) Is saturated

BIOLOGY

1) Match the column-I with column-II

	Column-I		**Column-II**
(I)	S phase	(i)	Most of organelles duplication

(II)	G_1 phase	(ii)	Tubuline proteins are synthesised
(III)	G_0 phase	(iii)	DNA Replication
(IV)	G_2 phase	(iv)	Cells are no longer proliferating

(1) I-ii, II-iii, III-iv, IV-i

(2) I-i, II-iv, III-ii, IV-iii

(3) I-iii, II-i, III-iv, IV-ii

(4) I-iii, II-ii, III-i, IV-iv

2)

Longest phase of meiosis is

(1) Prophase-I

(2) Prophase-II

(3) Anaphase-I

(4) Metaphase-II

3) Match item in column-I with column-II.

	Column-I		**Column-II**
(A)	Anaphase	(i)	Initiation of chromatin condensation
(B)	Telophase	(ii)	Chromatin condensation complete
(C)	Prophase	(iii)	Chromatid move to poles
(D)	Metaphase	(iv)	Nuclear envelope assembles (Reappear)

(1) A-i, B-ii, C-iii, D-iv

(2) A-ii, B-iv, C-i, D-iii

(3) A-iii, B-iv, C-i, D-ii

(4) A-iii, B-ii, C-i, D-iv

4) How many chromosome will the cell have during mitosis at G_1, S and in anaphase respectively if it has 14 chromosome prior to interphase :-

(1) 14, 28, 28

(2) 14, 14, 28

(3) 7, 7, 7

(4) 7, 14, 14

5)

The following statements describes prophase-I of meiosis :
(i) Thin thread like chromosomes appears
(ii) Appearance of recombination nodules
(iii) Formation of Bivalents/Tetrads
(iv) Terminalisation of chiasmata
(v) Appearance of chiasmata

Arrange the given statements in the correct sequence of their occurence during prophase-I :

(1) (i)→(iii)→(ii)→(v)→(iv)

(2) (i)→(ii)→(iii)→(iv)→(v)

(3) (i)→(iv)→(v)→(ii)→(iii)

(4) (i)→(iii)→(ii)→(iv)→(v)

6) Which one of the following stages is correctly described with its event.

(1)	Anaphase I	The bivalent chromosomes align on the equatorial plate
(2)	Diakinesis	Beginning of dissolution of synaptonemal complex.
(3)	Metaphase	Spindle fibres attach to kinetochores of chromosome
(4)	Zygotene	Apperance of recombination nodule

(1) 1

(2) 2

(3) 3

(4) 4

7) Which of the following is not correct for the anaphase in mitosis ?

(1) Centromeres split and chromatids separate.

(2) Daughter chromosomes move to opposite poles.

(3) Reformation of ER and Golgi apparatus.

(4) Number of chromosomes become double in cell.

8) Which one is the correct sequence of a cell cycle ?

(1) $G_1 \rightarrow$ Karyokinesis → cytokinesis → $G_2 \rightarrow S$

(2) $S \rightarrow G_2 \rightarrow$ Karyokinesis→ cytokinesis → prophase

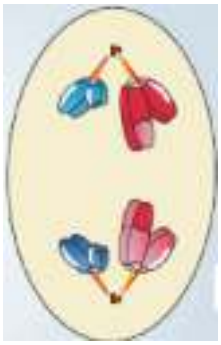
(3) $G_1 \rightarrow S \rightarrow G_2 \rightarrow$ Karyokinesis → cytokinesis

(4) Cytokinesis → Karyokinesis → $G_1 \rightarrow S \rightarrow G_2$

9) Zygotene is characterized by :-

(A) Pairing of homologous chromosomes
(B) Appearance of synaptonemal complex
(C) Formation of bivalents
(D) Formation of chiasmata

(1) A, B & D only
(2) B, C & D only
(3) All except C
(4) All except D

10) Identify the stage of cell cycle shown below :- 

(1) Anaphase-I
(2) Anaphase-II
(3) Telophase-I
(4) Telophase-II

11) Read the option carefully and find out the correct sequence :
(i) Synapsis is pairing of homologous chromosome.
(ii) Synthesis of RNA and Protein.
(iii) Action of enzyme Recombinase.
(iv) Centromeres do not separate but chromatids move towards opposite poles.

(1) i, ii, iii, iv
(2) iv, iii, i, ii
(3) iii, ii, iv, i
(4) ii, i, iii, iv

12) Which one is longest substage in oogenesis ?

(1) Zygotene
(2) Pachytene
(3) Diplotene
(4) Leptotene

13) **Statement-I :** In Telophase-I, chromosome do undergo some dispersion, they do not reach the extremely extend state of interphase nucleus.
Statetement-II : After telophase-II formation of tetrad of cells take place.

(1) Both statement-I and statement-II are incorrect

(2) Statement-I is correct but statement-II is incorrect

(3) Statement-I is incorrect but statement-II is correct

(4) Both statement-I and statement-II are correct

14) Some characteristics are given below :
Unicellular, Aquatic, Autotrophic and heterotrophic mode of nutrition and asexual reproduction by binary fission, Eukaryotic
Above mentioned characters are related to which kingdom ?

(1) Monera

(2) Protista

(3) Fungi

(4) Animalia

15) Match the following columns.

Column-I		**Column-II**	
A.	Chief producer in the oceans	1.	Diatoms
B.	Red tide	2.	Dinoflagellates
C.	Connecting link between plants and animals	3.	Euglenoids
D.	Fungus animals	4.	Slime moulds

(1) A-2,B-3,C-4,D-1

(2) A-1,B-2,C-3,D-4

(3) A-3,B-4,C-1,D-2

(4) A-4,B-1,C-2,D-3

16) Mycoplasma are classified under which of the following kingdoms ?

(1) Animalia

(2) Protista

(3) Monera

(4) Fungi

17) **Statement-I :-** Instead of a cell wall, Euglenoides have a protein rich layer called pellicle which make their body flexible.
Statement-II :- Under favourable conditions, slime moulds form an aggregation called plasmodium.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

18) Select the incorrect match :-

(1) Basidiomycetes - *Puffballs, Puccinia*

(2) Ascomycetes - *Penicillium, Aspergillus*

(3) Deuteromycetes - *Trichoderma, Alternaria*

(4) Phycomycetes - *Albugo, Colletotrichum*

19) Consider the following four statements (A-D) and select the option which includes all the **correct** ones only.
(A) Archaebacteria include methanogens halophiles and thermoacidophiles
(B) Bacteria as a group show the most extensive metabolic diversity
(C) Cyanobacteria have chlorophyll 'a' similar to green plants. (D) Mycoplasma can survive without oxygen.

(1) Statements B, C and D only

(2) Statements A, C only

(3) Statements B, D only

(4) Statements A,B,C and D

20) Read the following point :-
(A) Body septate and branched
(B) Some members are saprophytes or parasites
(C) Help in mineral cycling
(D) Reproduce by asexual spores known as conidia
Above points are related with which class of organisms

(1) Phycomycetes

(2) Euglenoids

(3) Deuteromycetes

(4) Basidiomycetes

21) Identify the correct match from the columns I, II and III :-

Column I		**Column II**		**Column III**	
1	*Albugo*	(a)	Ascomycetes	(i)	Obligate parasite
2	*Agaricus*	(b)	Deuteromycetes	(ii)	Decomposer of litter
3	*Aspergillus*	(c)	Phycomycetes	(iii)	Lab. Weeds
4	*Alternaria*	(d)	Basidiomycetes	(iv)	edible fungus

(1) 1-c-i, 2-a-iv, 3-a-iii, 4-b-ii

(2) 1-c-i, 2-d-iv, 3-d-iii, 4-a-ii

(3) 1-c-i, 2-d-iii, 3-d-iv, 4-b-ii

(4) 1-c-i, 2-d-iv, 3-a-iii, 4-b-ii

22) Which information is not related with lichen :-

(1) Show mutually useful association between two organisms.

(2) Algal component is known as phycobiont.

(3) They always grow in polluted areas.

(4) Fungi provides water in lichen.

23) Read the following statements and find out which one is incorrect :-

(1) Virus take over the machinery of the host cell to replicate themself

(2) Virus contain both RNA and DNA

(3) AIDS in human is caused by a virus

(4) Generally in plant viruses, ss RNA is present

24) Naked infectious RNA particle which cause disease in plants are :-

(1) Viroid

(2) Virus

(3) Prions

(4) Bacteria

25) *Mucor* and *Albugo* are members of :

(1) Phycomycetes

(2) Ascomycetes

(3) Deuteromycetes

(4) Basidiomycetes

26) Read the following statements and select the option with all correct statements.
(A) The cell walls of fungi are composed of chitin and polysacchrides
(B) *Aspergillus* is used extensively in biochemical and genetic work
(C) Large number of deuteromycetes are decomposers of litter and help in mineral cycling
(D) The sporophyte in moss is less elaborate than that in liver worts.

(1) A, C and D

(2) A and C

(3) A and D

(4) A, B, C, D

27) Read the following statement (A-D) and choose right option regarding them :-
(A) In Albugo, mycelium is aseprate and coenocytic
(B) Euglena are the chief producers in the oceans
(C) Cell wall of diatoms are indestructible due to deposition of silica
(D) True cell wall is absent in life-cycle of slime moulds

(1) (A), (B) wrong, (C), (D) Right

(2) (A), (C) wrong, (B), (D) Right

(3) (B), (D) wrong, (A), (C) Right

(4) (B), (C) wrong, (A), (D) Right

28) **Assertion :** Desmids is a member of protista.
Reason : It is an eukaryotic, unicellular, biflagellated, mostly marine and photosynthetic organism.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

29) Consider the following statements (A-D):-
(A) Fusion of two gametes which are similar in size and flagellated or non-flagellated is termed as isogamous.
(B) Fusion between one large, non motile female gamete and a smaller, motile male gamete is termed as Oogamous.
(C) Fusion of two gametes which are dissimilar in size is termed as anisogamous
(D) Both *Volvox* and *Fucus* show isogamous type of sexual reproduction.
Which of the above statement(s) is/are wrong?

(1) A, D

(2) B, C

(3) Only C

(4) Only D

30) Which one of the commercial product is used to grow microbes and in preparations of ice-creams and jellies :-

(1) Agar

(2) Algin

(3) Carrageen

(4) Pectose

31) The development of the zygote into young embryos takes place within the female gametophyte, this event is a precursor of the :-

(1) Seed habit

(2) Fruit habit

(3) Vascular habit

(4) Tissue habit

32) Prothallus is a gametophyte that occur in life cycle of -

(1) Pteridophytes

(2) Bryophyta

(3) Algae

(4) Angiosperm

33) Choose the incorrect statement from followings.
A. The most advanced gametophyte is found in mosses among bryophytes
B. Highly reduced gametophyte, which is having limited numbers of cells is found in flowering plants.
C. In both bryophytes and pteridophytes sporophyte is free living.
D. Gametophyte is always non-vascular.

(1) A and C

(2) C and D

(3) B and D

(4) Only C

34) Match the following columns :

Column-I		**Column-II**	
A	Green algae	i	*Dryopteris*
B	Lycopsida	ii	*Equisetum*
C	Horse tail	iii	*Selaginella*
D	Pteropsida	iv	*Ulothrix*

(1) A → iv; B → iii; C → ii; D → i

(2) A → iii; B → ii; C → i; D → iv

(3) A → ii; B → i; C → iv; D → iii

(4) A → i; B → iv; C → iii; D → ii

35) Consider the following four statements (a-d) and select the option which include all the correct ones only :-
(a) Unlike bryophytes and pteridophytes, in gymnosperms the male and female gametophyte do not have an independent free living existence.
(b) In some pteridophytes, the female gametophyte is retained on the parent sporophyte for variable period.
(c) Zygote develops into embryo, retained with in female gametophyte in some pteridophytes.
(d) Bryophytes do not play any role in plant succesion occuring on bare rock or soil or terrestrial habitate.

(1) b, c and d

(2) a, c and d

(3) a, b and d

(4) a, b and c

36) Which of the following statement is incorrect about Gymnosperm.

(1) Pollination takes place by air in gymnosperms.

(2) They are heterosporous

(3) Before and after fertilization, ovules are not enclosed by any ovary wall and remain exposed.

(4) All Gymnosperm are dioecious without any exception.

37) In pinus endosperm cells have 15 chromosomes then how many chromosomes are present in the sieve cells :-

(1) 15

(2) 30

(3) 45

(4) 10

38)

Read the following statements :
Statement A: Bryophytes live in soil but depend on water for sexual reproduction.
Statement B: The male gametes (sperm) of bryophytes are biflagellated.

Choose the correct option :-

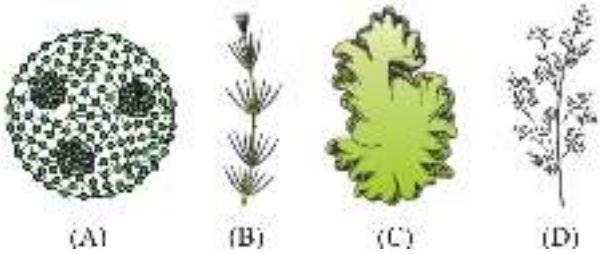
(1) Both the statements A and B are incorrect

(2) Both the statements A and B are correct

(3) The statement A is correct but B is incorrect

(4) The statement A is incorrect but B is correct

39) Examin the figure A, B, C and D. In which one of the four options all the items A, B, C and D are correct ?



	A	B	C	D
(1)	Volvox	Chara	Fucus	Porphyra
(2)	Volvox	Chara	Porphyra	Polysiphonia
(3)	Volvox	Chara	Polysiphonia	Porphyra
(4)	Chara	Volvox	Porphyra	Polysiphonia

(1) 1

(2) 2

(3) 3

(4) 4

40) Which of the following is lacking in gymnosperms ?

(1) Ovary

(2) Embryo

(3) Seeds

(4) Embryosac

41) Which of the following statement is correct ?

(1) In ferns asexual reproduction takes place by green, multicellular buds called gemmae.

(2) The sporophyte in bryophytes is differentiated into foot, seta and capsule and it is a free living generation

(3) Protonema is a green, creeping, branched and filamentous stage in mosses, which develops directly from a spore.

(4) The liverworts have peristomial teeth for spore dispersal.

42) Which one of the following is example of homosporous and heterosporus respectively ?

(1) *Azolla, Pinus*

(2) *Sellaginella, Adiantum*

(3) *Dryopteris, Cycas*

(4) *Equisetum, Sphagnum*

43) The sporophyte of the mosses is composed of:-

(1) Only capsule

(2) Foot and capsule

(3) Foot, seta and capsule

(4) Only foot

44) Zoospore of brown algae have :-

(1) Two equal flagella on tip

(2) Two unequal flagella on tip

(3) Two laterally placed equal flagella

(4) Two laterally placed unequal flagella

45) Match Column-I with Column-II and select the correct option from given blow.

	Column-I		**Column-II**
(A)	Psilopsida	(I)	*Psilotam*
(B)	Lycopsida	(II)	*Equisetum*
(C)	Sphenopsida	(III)	*Selaginela*
(D)	Pteropsida	(IV)	*Dryopteris*

(1) A - I B - II C - III D - IV

(2) A - IV B - III C - II D - I

(3) A - I B - III C - II D - IV

(4) A - III B - IV C - I D - II

46) Consider the following four statements whether they are correct or wrong -
(a) Mosses have dominant and independent sporophyte
(b) Origin of seed habit can be traced in pteridophytes
(c) In gymnosperms male and female gametophytes are present within sporangia located in sporophyte
(d) Algin is obtained from red algae and carrageen from brown algae.
The two wrong statements together are :-

(1) Statements (a) and (b)

(2) Statements (b) and (c)

(3) Statements (b) and (d)

(4) Statements (a) and (d)

47) **Statement-I :-** The sporophyte in moss is more elaborate than that in liver worts.
Statement-II :- The spore dispersal of liverwort is more elaborate than mass.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

48) **Assertion :** In *Marchantia*, gametophyte stage consists of two stages protonema and leafy stage.
Reason : Gametophyte of *Funaria* is more developed than gametophyte of *Marchantia*.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is False but the Reason is True.

(4) Both Assertion & Reason are False.

49) How many characters are related to gymnosperm ?
(a) In cycus male cone and megasporophyll are born on different tree.
(b) Development of pollen grain takes place in microsporophyll
(c) Gymnosperms produce haploid microspore and megaspores.
(d) In gymnosperms seeds are covered.
(e) Pollen grain is released from the microsporangium.

(1) a, b, c, e

(2) b, c, d, e

(3) a, b, c, d, e

(4) b, c, d

50) Read the following statements regarding ribosomes :
(i) Smallest membrane bound organelle.
(ii) They are composed of RNA and proteins.
(iii) In eukaryotes 70S ribosomes are present in mitochondria.
(iv) In prokaryotes, 70S ribosomes are associated with plasma membrane.

(v) All the components of ribosome are synthesised in nucleolus.
Which of the above statements are incorrect?

(1) (i), (iii) and (v)

(2) (i) and (v)

(3) Only (v)

(4) (ii), (iii), (iv) and (v)

51) Choose the correct option for cell organelles which are the part of endomembrane system:
(i) Endoplasmic Reticulum (ii) Vacuoles
(iii) Mitochondria (iv) Plastids
(v) Peroxisome (vi) Golgibody
(vii) Lysosome

(1) i, ii, v, vii

(2) ii, iv, vi, vii

(3) i, ii, vi, vii

(4) i, iii, vi, vii

52) An elaborate network of filamentous.....structure present in the cytoplasm is collectively referred to as the..... :-

(1) Cytoskeleton, Proteinaceous

(2) Proteinaceous, Cytoskeleton

(3) Lipoidal, Mitochondria

(4) Lipoidal, Plasma membrane

53) All prokaryotes have a cell wall surrounding the cell membrane except in

(1) Bacteria

(2) Plant

(3) Fungi

(4) Mycoplasma

54) Which of following cell organelle divides intracellular space into two distinct compartments i.e. Luminal & extra-luminal compartments

(1) Endoplasmic reticulum

(2) Mitochondria

(3) Lysosome

(4) Peroxisome

55) Match the column

	Column-I		**Column-II**
(a)	Rough endoplasmic reticulum	(i)	Lipid

(b)	Smooth endoplasmic reticulum	(ii)	Cell secretion
(c)	Golgi apparatus	(iii)	Protein
(d)	Cytoskeleton	(iv)	Maintenance of cell shape

(1) a-iii, b-ii, c-i, d-iv

(2) a-iii, b-i, c-ii, d-iv

(3) a-iii, b-i, c-iv, d-ii

(4) a-iv, b-iii, c-ii, d-i

56) Which of the following is not the function of cell wall ?
(i) Provides shape to the cell.
(ii) Protects the cell from mechanical damage and infection.
(iii) Helps in cell to cell connection.
(iv) Provides barrier to undesirable macromolecules.
(v) Helps in cell furrow formation

(1) only (iii)

(2) only (iv)

(3) only (ii), (iii) and (v)

(4) only (v)



57) Which of the following pair is correct for given diagram ?

	Diagram is	**Involve in**
(1)	Nuclear membrane	DNA Synthesis
(2)	E.R.	Steroid Synthesis
(3)	Golgi complex	Glycoprotein formation
(4)	E.R.	Protein synthesis

(1) 1

(2) 2

(3) 3

(4) 4

58) Match the following and choose the **correct** option :-

List-I	**List-II**

a	Cell wall	i	Protein synthesis
b	Cell membrane	ii	Hydrolytic enzyme
c	SER	iii	Lipid synthesis
d	RER	iv	Cellulose
e	Lysosome	v	Quasifluid nature

(1) a = v b = iv c = iii d = i e = ii

(2) a = i b = ii c = iii d = iv e = v

(3) a = iv b = v c = i d = iii e = ii

(4) a = iv b = v c = iii d = i e = ii

59) **Assertion:-** Centrosome is an organelle usually containing two cylindrical structure called centrioles.
Reason :- Both centriole in a centrosome lie parallel to each other in which each has an organisation like the cart wheel.

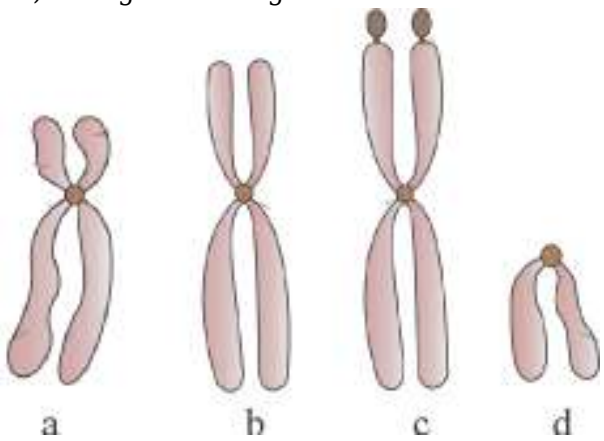
(1) Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Assertion and Reason are true but Reason is not a correct explanation of the Assertion.

(3) Assertion is true but the Reason is false.

(4) Both Assertion and Reason are false.

60) Recognise the figure and find out the correct matching



(1) b-acrocentric, a-telocentric, d-metacentric, c-submetacentric

(2) a-acrocentric, d-telocentric, c-metacentric, b-submetacentric

(3) a-acrocentric, d-telocentric, b-metacentric, c-submetacentric

(4) d-acrocentric, c-telocentric, a-metacentric, b-submetacentric

61) The mycelium of Basidiomycetes is -

(1) Unbranched and aseptate

(2) Unbranched and septate
(3) Branched and aseptate
(4) Branched and septate

62) In which of the following the cell wall stiff cellulose plate on the outer surface ?

(1) Diatoms
(2) Dino flagellates
(3) Euglenoid
(4) Desmids

63) Match the following column-I with column-II.

	Column-I		**Column-II**
A	Diatoms	(a)	Outermost layer in pellicle
B	Dino flagellates	(b)	Saprophytic protists
C	Euglinoids	(c)	Red tide
D	Slime moulds	(d)	Cell wall embedded with silica

(1) A-d, B-c, C-a, D-b
(2) A-d, B-c, C-b, D-a
(3) A-d, B-a, C-b, D-c
(4) A-d, B-b, C-a, D-c

64) The Female sex organ of bryophytes is called ___A___ which is flask shaped and produces ___B___ .

(1) A – Archegonium, B – Two eggs
(2) A – Antheridium, B – One egg
(3) A – Archegonium, B – One egg
(4) A – Antheridium, B – Many antherozoids

65) Which of the following statement is not correct ?

(1) Algin is obtained from brown algae.
(2) Majority of red algae are marine.
(3) The red thalli of most red algae are unicellular.
(4) Polysiphonia and porphyra belong to red algae.

66) Which one of the following is not include in kingdom protista ?

(1) Anabaena
(2) Paramoecium
(3) Golden algae
(4) Gonyaulax

67) **Assertion (A) :** In gymnosperms male and female gametophytes do not have an independent free living existance.
Reason (R) : In gymnosperms germination of microspore and megaspore occur inside microsporangium and megasporangium respectively present on sporophyte.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Only (A) is true.
(3) Both (A) and (R) are false.
(4) Both (A) and (R) are true and (R) is not the correct explanation of (A).

68) Select the correct statement -

(1) All fungi are parasitic
(2) All fungi are saprophytic
(3) All fungi are heterotrophic
(4) All fungi are symbionts

69) Read all statement carefully regarding pteridophyte and select incorrect statement.

(1) Mostly pteridophytes and homosporous.
(2) Precursor to the seed habit considered an important step in evolution in homosporous pteridophytes.
(3) Pteridophytes are first vascular plant.
(4) Pteridophytes are used for medicinal purposes and soil-binders.

70) Select the incorrect match -

(1) *Rhizopus* – Bread mould
(2) *Ustilago* – Smut fungus
(3) *Puccinia* – Rust fungus
(4) *Agaricus* – Imperfect fungi

71) Gymnosperms differs from pteridophytes in -

(1) Having fruits
(2) Having naked ovule
(3) Having flower
(4) Having antheridium

72) Vegitative reproduction in mosses is by _____ and _____ in the secondary protonema.

(1) Spore formation, budding
(2) Gemmae, budding
(3) Fragmentation, budding
(4) Fragmentation, spore formation

73) Which of the following statement are correct ?

(A) Bryophytes are called amphibians of the plant kingdom.
(B) Main plant body of the bryophytes is haploid.
(C) Protonema stage in moss develops directly from a spore.
(D) In Bryophyte, sporophyte is free living and photosynthetic.
Choose the most appropriate answer from the options given below.

(1) A and D only

(2) A, B and C only

(3) A, B and D only

(4) B and D only

74) **Statement-I :** All protozoan are heterotrophs and live as predators or parasites.
Statement-II : Chrysophytes are found in fresh water as well as in marine water.

(1) Both statement-I and statement-II are correct.

(2) Statement-I is incorrect but statement-II is correct.

(3) Statement-I is correct but statement-II is incorrect.

(4) Both statement-I and statement-II are incorrect.

75) In which of the following plant groups the female reproductive organ in archigonium ?

(1) Bryophyta, Pteridophyta, Gymnosperm

(2) Algae, Bryophyta, Pteridophyta

(3) Algae, Bryophyta, Gymnosperm

(4) Pteridophyta, Gymnosperm, Angiosperm

76) Select the incorrect pair :

(1)	Liverworts	Marchantia
(2)	Mosses	Sphagnum
(3)	Pteridophyte	Fern
(4)	Gymnosperm	Selaginella

(1) 1

(2) 2

(3) 3

(4) 4

77) How many statement are correct for *Cycas* ?
(a) *Cycas* in the tallest gymnosperm.
(b) The ovule are not enclosed by any ovary wall and remain exposed, both before and after fertilisation.
(c) In *Cycas* the pinnate leaves persist for a few years.
(d) The development of zygote into embryo takes place with in the female gametophyte.

(1) Two

(2) Three

(3) One

(4) Four

78) Match column-I with column-II and choose the correct option :

	Column-I		**Column-II**
(A)	*Polysiphonia*	(i)	Hetrosporous nature
(B)	*Marchantia*	(ii)	Non-motile gametes
(C)	*Selaginella*	(iii)	Dioecious gametophytes
(D)	*Cycas*	(iv)	Coralloid roots

(1) A-ii, B-i, C-iv, D-iii

(2) A-iii, B-i, C-ii, D-iv

(3) A-iv, B-ii, C-iii, D-i

(4) A-ii, B-iii, C-i, D-iv

79) Read the following statement (A to E) carefully :
(A) The pteridophytes include horstails and mosses.
(B) *Selaginella* is a microphyllous plant.
(C) The gymnosperms are heterosporous.
(D) In *Pinus* and *Cedrus* the stem is branched.
(E) In *Selaginella* sporophylls may form distinct compact structure called strobili or cones.
How many of the above statement are not correct.

(1) Two

(2) Three

(3) Four

(4) One

80) Select the incorrect match.

(1) Ferns – protonema

(2) Marchantia – Gemmae

(3) Pinus – mycorrhiza

(4) Gelidium – Agar

81) Read the column-I and column-II and identify their correct match.

	Column-I		**Column-II**
(A)	Glycocalyx	(i)	Gas vacuoles
(B)	Mesosome	(ii)	Slime layer or capsule
(C)	Chromatophores	(iii)	Cell wall formation
(D)	Inclusion bodies	(iv)	Pigments

		(v)	Sap vacuole
		(vi)	Chloroplast
		(vii)	Ribosome

(1) A-iii, B-iv, C-i, D-ii

(2) A-i, B-iii, C-ii, D-iv

(3) A-iv, B-i, C-ii, D-iii

(4) A-ii, B-iii, C-iv, D-i

82) Read the following statement -
(a) In gymnosperms coralloid roots are associated with N_2-fixing cyanobacteria.
(b) Gymnosperms produce both microspores and megaspores.
(c) Pteridophytes are the first terrestrial plants to possess vascular tissue-xylem and phloem.
(d) The dominant phase in the life cycle of pteridophytes is the gametophytic plant body.
(e) Pteridophytes are found in cool, damp, shady places but some flourish well in sandy-soil conditions also.
Choose the correct statement :

(1) a, b, d & e

(2) a, c, d & e

(3) a, b, c, & d

(4) a, b, c & e

83) Match the column-I and column-II and choose the correct option.

	Column-I		**Column-II**
(i)	Coralloid root	(a)	*Pinus*
(ii)	Mycorrhizal root	(b)	*Polytrichnum*
(iii)	Red wood tree	(c)	*Cycas*
(iv)	Moss	(d)	*Marchantia*
(v)	Dioecious bryophyte	(e)	*Sequoia*

(1) i-c, ii-e, iii-d, iv-a, v-b

(2) i-c, ii-a, iii-d, iv-b, v-e

(3) i-c, ii-a, iii-e, iv-b, v-d

(4) i-c, ii-a, iii-b, iv-e, v-d

84) How many of the following members belong to phaeophyceae ?
Fucus, Porphyra, Chara, Sargassum, Gelidium, Dictyota, Ectocarpus
Choose the correct option.

(1) Five

(2) Four

(3) Three

(4) Two

85) Given below are the stages in the life cycle of Funaria arrange the following stages in the correct sequence.
(A) Spore germination
(B) Meiosis in spore mother cells
(C) Fertilisation
(D) Formation of archegonia & antheridia in gametophyte.
(E) Transfer of antherozoids to the archegonia in presence of water.
Choose the correct answer from the options given below :

(1) A → D → C → B → E
(2) C → D → A → B → E
(3) E → C → A → E → B
(4) B → A → D → E → C

86) While studying the structure of a newly discovered bryophyte, a researcher observed that the plant body is thalloid lacks true roots stems and leaves and the sporophyte is completely dependent on the gametophyte for nutrition which characteristic feature of bryophytes does this observation indicate ?

(1) Dominance of sporophyte generation
(2) Independent sporophyte
(3) Dominance of gametophyte generation
(4) Presence of vascular tissues

87) Sporophytic plant body of *Pinus* have -

(1) Male and female cone found on same plant body.
(2) Unbranched stem
(3) Coralloid root
(4) All of the above

88) Which of the following disease is caused by viruses ?

(1) Mad cow disease in cattle
(2) Cr-Jakob disease
(3) Small pox
(4) Potato spindle tuber disease

89) **Assertion (A) :** In the five kingdom classification of whittaker there is no mention of viruses.
Reason (R) : Viruses are not considered truly 'living' as they are noncellular.

(1) (A) is true but (R) is false
(2) Both (A) and (R) true & (R) is correct explanation of (A)
(3) (A) is false but (R) is true
(4) Both (A) and (R) true but (R) is not correct explanation of (A)

90) Which of the following are the characters of dinoflagellates ?

(a) Mostly marine and photosynthetic
(b) Appear yellow, green, brown, blue or red in colour
(c) Biflagellated organisms without cell wall
(d) Parasitic multicellular forms

(1) a and b only
(2) c and d only
(3) b, c and d only
(4) a, b and d only

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	2	4	3	2	2	2	3	1	4	1	3	3	2	1	3	4	2	1	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	1	4	3	4	3	2	2	2	3	3	4	3	2	3	3	3	1	3	1
Q.	41	42	43	44	45															
A.	2	2	1	2	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	3	3	3	3	2	4	4	4	2	4	4	3	2	1	1	3	4	2	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	4	3	4	2	1	4	2	1	2	1	2	3	1	3	2	3	2	3	1
Q.	86	87	88	89	90															
A.	4	3	3	4	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	1	3	2	1	3	3	3	4	1	4	3	4	2	2	3	4	4	4	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	3	2	1	1	2	3	3	4	1	1	1	4	1	4	4	2	2	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	3	4	3	4	2	3	1	2	3	2	4	1	2	4	3	4	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	2	1	3	3	1	1	3	2	4	2	3	2	1	1	4	2	4	4	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	3	2	4	3	1	3	2	1										

SOLUTIONS

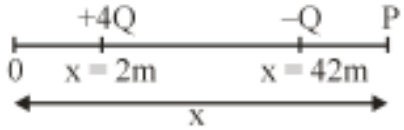
PHYSICS

1)

Question Explanation : We need to locate the point where the 3rd charge +Q should be placed so that it remains in the state of equilibrium, due to other given charges.

Concept : Equilibrium in case of discrete charge systems.

Solution :



For F = 0, E = 0 for point P-

$$\frac{K(4Q)}{(x-2)^2} = \frac{KQ}{(x-42)^2}$$

taking square root

$$\frac{2}{x-2} = \frac{1}{x-42}$$

$2x - 84 = x - 2$

$x = 82$ m

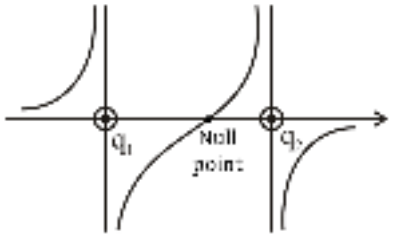
Final answer : Option (4)

2)

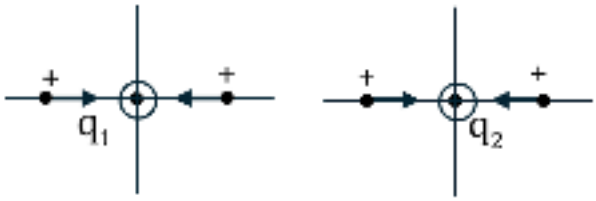
Explain Question : From the given graph (Electric field verses distance) we need to select the correct statement about signs and magnitudes of the charge q_1 and q_2.

Concept : Electric field intensity due to discrete system of charges.

Solution :

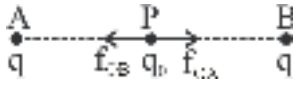


There is Null point ⇒ (Both are like charges)
the Null point is close to q_2 ⇒ q_2 is weak charge (⇒ $|q_1| > |q_2|$)

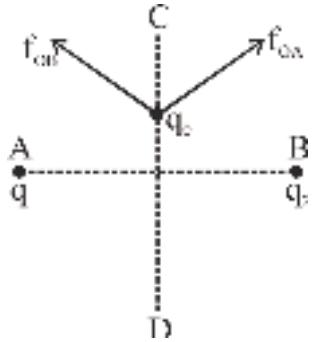


Due to the forces on the positive test charges, q_1 and q_2 both must be negative.

Final Answer : option (2)

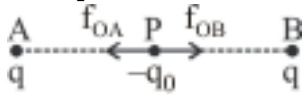
3) **case I :** 

$f_{OB} = f_{OA}$

when q_0 is displaced slightly towards charge q at B, $f'_{OB} > f'_{OA}$ [Hence, stable equilibrium] **case II :** 

$\vec{f}_{OA} + \vec{f}_{OB}$ is along DC towards C

⇒ Unstable equilibrium

case III : 

$f_{OA} = f_{OB}$ (when '$-q_0$' is at 'P')

If $-q_0$ is displaced slightly towards B,

$f'_{OB} > f'_{OA}$

⇒ Unstable equilibrium

4) **Explain Question:**

Question asks you the net electric field at a center of a thin semi - circular ring of radius r.

Concept:

Electric field at a point due to continuous charge distribution:-

Solution:

Electric field of uniformly charged arc at center,

$$|\vec{E}| = \frac{2k\lambda}{r}\sin\left(\frac{\phi}{2}\right) = \frac{2k\lambda}{r}\sin\left(\frac{180^\circ}{2}\right)$$

$$\left|\vec{E}\right| = \frac{2k\lambda}{r}\sin 90^\circ = \frac{2k\lambda}{r} \quad \ldots. (1)$$

Here, $\lambda = \frac{q}{L}$

For semi - circular ring

$L = \pi r$

$\lambda = \frac{q}{\pi r}$

So, $\left|\vec{E}\right| = \frac{2k\lambda}{r}$

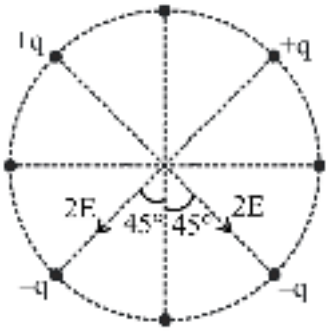
$$= 2 \times \frac{1}{4\pi\varepsilon_0} \times \frac{q}{\pi r} \times \frac{1}{r}$$

$$\left|\vec{E}\right| = \frac{q}{2\pi^2\varepsilon_0 r^2}$$
$$\vec{E} = -\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$$

Final Answer: Option (3)

5)

Electric field at centre due to single charge
$$E = \frac{kq}{r^2}$$
If like charge are at point located diametrically opposite then their cancel electric field each other. So field is due to remaining charge.

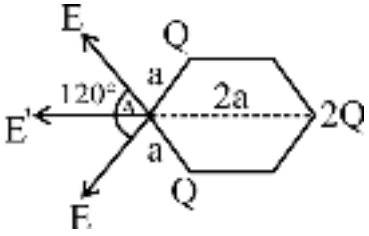


$$E_{net} = 2\sqrt{2}E = 2\sqrt{2}\frac{kq}{r^2}$$

6)

Explain Question : We are to find how many electrons must be removed from a silver coin so that it acquires a +2.4 C charge.
Concept : This question is based on
Each electron carries a charge e = 1.6×10^{-19} C.
Number of electrons removed : n = $\frac{Q}{e}$.
Solution :
$$n = \frac{2.4}{1.6 \times 10^{-19}} = 1.5 \times 10^{19}$$
Final answer : 2

7) **Explain the Question :** Find the net electric field at vertex
Concept : This question is based on
Solution :



where $E = \frac{KQ}{a^2}$ & $E' = \frac{K2Q}{4a^2}$

Resultant of E & E (which are at 120°) is E and parallel to E'

$E_{Net} = E + E'$

$= \frac{KQ}{a^2} + \frac{K(2Q)}{4a^2}$

$= \frac{3}{2}\frac{KQ}{a^2}$

Final Answer : (2)

8)

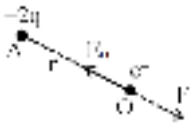
All combinations are based on following properties of electric field lines.
Tangent to electric field line represent the direction of resultant electric field at a point.
Field lines originate from a positive charge and terminate on negative charge.
Number of field lines associated with a charge is directly proportional to the magnitude of charge.
Density of field lines is directly proportional to the strength of electric field.

9) $Q_{system} = 18 + 22 = 40$ C

$\frac{Q_1'}{Q_2'} = \frac{10}{30} = \frac{1}{3}$

$\Rightarrow Q_1' = 10C \; Q_2' = 30C$

So + 8 C charge will flow from A to B.

10) 

$E_0 = \frac{2kq}{r^2}$ along $\overrightarrow{OA}$

Force on electron, $F = \frac{2kq}{r^2}$ along $\overrightarrow{AO}$

11) $F_e = mg \Rightarrow q = mg$

$\Rightarrow 10^{-6} \times E = 10^{-6} \times 10$

$\Rightarrow E = 10$ V/m

12) $\frac{q}{m} = \frac{1.6 \times 10^{-19}}{9.1 \times 10^{-31}} = 1.8 \times 10^{11}$ C/kg

13) $x = \sqrt{3}R$

$E = \frac{kQx}{(R^2 + x^2)^{3/2}}$

$$E = \frac{kQ}{(R^2 + 3R^2)^{3/2}}\sqrt{3}R$$

$$E = \frac{kQ}{8R^3}\sqrt{3}R = \frac{\sqrt{3}}{8}\frac{kQ}{R^2}$$

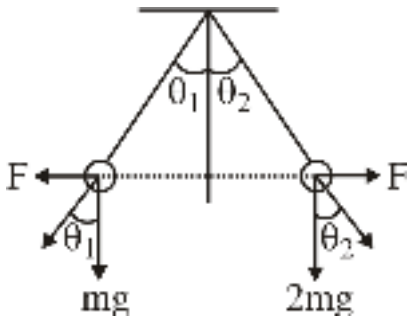
14)

$Q = ne$

$$n = \frac{Q}{e} = \frac{1}{1.6 \times 10^{-19}}$$

$= 6.25 \times 10^{18}$ es

15)



It is clear from figure that θ_1 is greater than θ_2.

16)

$U = x^3 + 3x^2 - 9x + 6$

$$\frac{dU}{dx} = 3x^2 + 6x - 9 = 0$$

$x^2 + 2x - 3 = 0$

$x^2 + 3x - x - 3 = 0$

$x(x + 3) - 1(x + 3) = 0$

$x = 1, \quad x = -3$

$$\boxed{\frac{d^2U}{dx^2} = 6x + 6}$$

So at $x = 1$

$$\frac{d^2U}{dx^2} = +ve$$

$x = 1$ is minima

So 'U' is minimum at $x = 1$

$U_{min} = 1^3 + 3(1)^2 - 9(1) + 6$

$= 1 + 3 - 9 + 6 = 1$ J

17) $$S_\infty = \frac{a}{1 - r}$$

$$S_\infty = \frac{1}{1 - \left(\frac{-1}{3}\right)} = \frac{3}{4}$$

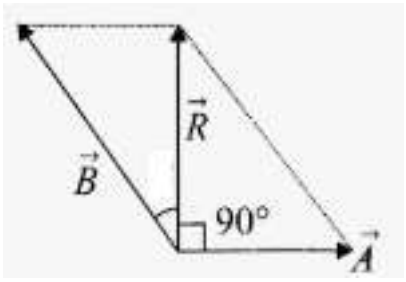
18) $y = 3x + 5 \quad \Rightarrow \quad \frac{dy}{dt} = 3\frac{dx}{dt}$
⇒ y–coordinate changes at faster rate.

19) x = (sin2°) (cos2°)
$x \simeq (\sin 2°)\ (1) = \sin\left(\frac{2\pi}{180}\right) \simeq \left(\frac{\pi}{90}\right)$

20) Graph of (cos x) at line y = +1

21)

Explanation :- Acceleration of object due to 5 forces is given when F_1 removed then acc. Changes, we have to find acceleration due to removed force.
Concept :- Newton's laws of motion.
Solution :-
Let mass of particle is m
$$\frac{\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5}{m} = 4\hat{i}$$
$$\frac{\vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5}{m} = 7\hat{j}$$
$$\frac{\vec{F}_1}{m} + 7\hat{j} = 4\hat{i}$$
$$\frac{\vec{F}_1}{m} = 4\hat{i} - 7\hat{j}$$
$\frac{\vec{F}_1}{m}$ ⇒ acceleration due to
$$\left|\frac{\vec{F}_1}{m}\right| = \sqrt{4^4 + 7^2}$$
$= \sqrt{65}$ m/sec^2
Final answer :- $\sqrt{65}$ m/sec^2

22) 

$$\tan 90° = \infty = \frac{1}{0} = \frac{B\sin\theta}{A + B\cos\theta}$$
A + B cos θ = 0
cos θ = -A/B
cos θ = -1/2 = θ = 120°

23) $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$

$\vec{a}+\vec{b}-\vec{c}=0$
$\vec{a}+\vec{b}=\vec{c}$
Resultant of 2 unit vector is also a unit vector
$\Rightarrow \theta = 120^{\circ}$

24)

Component of $\vec{P}$ along x axis is 10 cos 37° = 8 to balance $\vec{Q}$ is also 8 along negative x-axis. then $\vec{P}+\vec{Q}$ will be along y axis.

25)

Angle between two vector

$$\cos\theta = \frac{\vec{A}\cdot\vec{B}}{AB} = \frac{1}{\left(\sqrt{3}\right)(1)}$$

$$\cos\theta = \frac{1}{\sqrt{3}}$$

26) $\vec{\tau} = \vec{r}\times\vec{F}$

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & -1 \\ 1 & 1 & -1 \end{vmatrix}$$

$\vec{\tau} = -3\hat{i}+2\hat{j}-\hat{k}$

27)

u cos θ = v cos ϕ

28) $\dfrac{\frac{1}{2}mU_{x_1}^2}{\frac{1}{2}mU_{x_2}^2} = \dfrac{4}{1} \Rightarrow \dfrac{U_{x_1}}{U_{x_2}} = \dfrac{2}{1}$

$$\frac{\frac{U_{y_1}^2}{2g}}{\frac{U_{y_2}^2}{2g}} = \frac{4}{1} \Rightarrow \frac{U_{y_1}}{U_{y_2}} = \frac{2}{1}$$

$$R = \frac{2U_xU_y}{g} \Rightarrow \frac{R_1}{R_2} = \frac{4}{1}$$

29)

Explain Question : In this question two projectiles are fixed simullaneously. One from point A at angle 60° with speed μ, and another from point B (10 m away) at 20m/s. We need to find μ, so mal may collide mid-air.
Concept : This question is based on collision of projectile with other projectile.

Solution : For collision.
1. Their x-coordinate must be equal at same time 't'.
2. Their y-coordinate must be equal at same time 't'.

Solving for vertical displacement.
⇒ Vertical component of initial velocity as A = μ sin 60°
⇒ Vertical component of initial velocity as B = 20 sin 45°
For collision :
⇒ μ sin 60° = 20 sin 45°

$$\Rightarrow \mu \times \frac{\sqrt{3}}{2} = 20 \times \frac{1}{\sqrt{2}} \left(\sin 60^\circ \frac{\sqrt{3}}{2} \sin 45^\circ \frac{1}{\sqrt{2}} \right)$$

$$\Rightarrow \mu = \frac{20}{\sqrt{2}} \times \frac{2}{\sqrt{3}} = 20 \times \frac{\sqrt{2}}{\sqrt{3}}$$

$$\Rightarrow \mu = \frac{20 \times \sqrt{2}}{3}$$

Final answer : 2

30) **Explain the Question :** Two balls, A and B, are thrown from a tower of height H. Ball A is thrown with $10\sqrt{2}$ m/s at 45° upward from the horizontal, and Ball B is thrown with 20 m/s at 30° upward from the horizontal. determine which ball hits the ground first.
Concept : This question is based on projectile motion - oblique projection.
Formula :

$$s_y = u_y t + \frac{1}{2} g t^2$$

Solution : The **time of flight (t)** for a projectile to hit the ground is governed **only by the vertical motion**. Since both balls start at the same height H and undergo the same constant vertical acceleration (g), the time of flight depends only on their initial vertical velocity component (u_y).
1. **Ball A**. $U_A = 10\sqrt{2}$ m/s at $\theta_A = 45°$ **upward**.

$u_{yA} = U_A \sin(\theta_A) = 10\sqrt{2} \sin(45°) = 10\sqrt{2}\left(\frac{1}{\sqrt{2}}\right) = +10$m/s
2. **Ball B** : $U_B = 20$m/s at $\theta_B = 30°$ **upward**

$u_{yB} = U_B \sin(\theta_B) = 20 \sin(30°) = 20\left(\frac{1}{2}\right) = +10$m/s
Since the initial vertical velocities are identical ($u_{yA} = u_{yB} = 10$ m/s) and they both start from the same height H, the time of flight (t) will be the same for both balls, as shown by the quadratic equation for t:

$$H = -(10)t + \frac{1}{2} g t^2$$

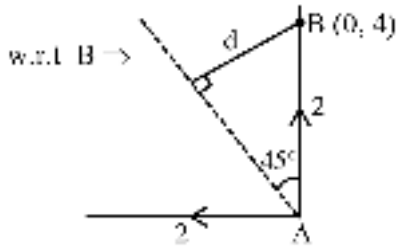
Therefore, both balls will strike the ground at the same time.
Final answer : 3

31)

Explain Question : Two particles moves at different position in different directions we need to find the minimum separation between then & time.

Concept : Relative motion in a plane.

Solution :



d = 4cos45°

d = $2\sqrt{2}$ m

t = $\frac{d}{V_r} = \frac{2\sqrt{2}}{2\sqrt{2}} = 1 \text{ sec.}$

Final Answer : option (3)

32) **Explain the Question :**

We have to find velocity of man such that rain drop hit him with speed of 5 m/s

Concept : This question is based on rain-man problem.

Solution :

Let velocity of man with respect to ground = $v\hat{i}$ Velocity of rain w.r.t. man $(\vec{v}_r) = (2\hat{i} - 4\hat{j})$

$\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$

$|\vec{v}_{rm}| = (2\hat{i} - 4\hat{j}) - v\hat{i}$

$|\vec{v}_{rm}| = [2\hat{i} - v\hat{j} - 4\hat{j}]$

$5 = \sqrt{(2-v)^2 + (4)^2}$

$2s = (2 - v)^2 + 16$

$(2v)^2 = 9$

2–v = ±3

2 – v = ± 3

v = $5\hat{i}$ and $-1\hat{i}$

Final answer : 4

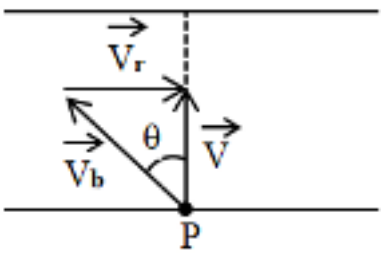
33) For equation of path

t = $\frac{x}{\alpha}$ put in another equation

$\Rightarrow y = \alpha.\frac{x}{\alpha}\left(1 - \beta.\frac{x}{\alpha}\right)$

$\Rightarrow y = x\left(1 - \frac{\beta}{\alpha}x\right)$

34)



$\sin\theta = \frac{v_r}{v_b} = \frac{4}{10}$

$\theta = \sin^{-1}\left(\frac{2}{5}\right)$ from PQ upstream

35) Time of flight = $t_1 + t_2$

= 1 + 3 = 4sec

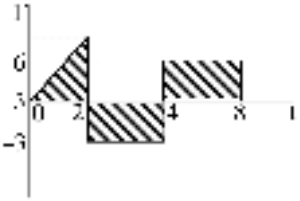
$$\frac{2u\sin 30}{g} = 4$$

u = 40 m/sec

36)

Mass is measurement of inertia

37) **Explain the Question :** Find change in momentum form 0 to 8sec.
Concept : This question is based on change in momentum (F vs t curve).
Solution :



Area under F-t curve gives change in momentum (ΔP)

$$\Delta P = \int F dt$$

$$\Delta P = \frac{1}{2} \times 6 \times 2 + (-3)(4-2) + 3 \times (8-4)$$

ΔP = 6 – 6 + 12
ΔP = 12 Ns
Final Answer : (3)

38)

Correct option is (1)

39) $F = \frac{(mV)n}{\Delta t}$

$144 = (40 \times 10^{-3})(1200)\frac{n}{\Delta t}$

$\frac{n}{\Delta t}$ = 3 bullets/sec.

40) **Question Explanation:**
Calculating contact forces between three blocks.

Concept:
System acceleration and Free Body Diagrams.

Solution:
Total mass = 3 + 2 + 1 = 6kg
Acceleration (a) = $\frac{12N}{6kg}$ = 2m/s^2
f_2 (force on 3kg block) = m^3 × a = 3 × 2 = 6N
f_1 (force between 2kg and 1kg) pushes (3kg + 2kg)
f_1 = (3 + 2) × a = 5 × 2 = 10N

Final Answer: Option (1)

41)

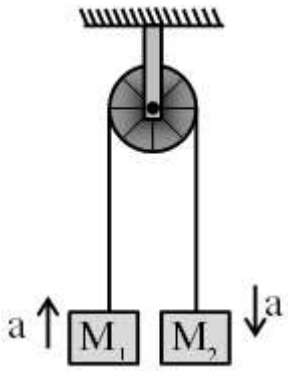
T sin θ = F ⇒ T = $\frac{F}{\sin\theta}$

42) **Explain Question:**
Question ask to find the acceleration of the 4 kg block & tension in string

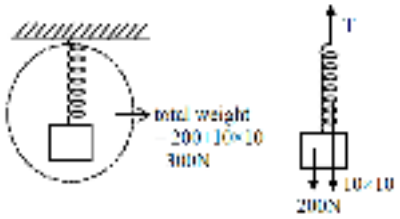
Concept:
Motion of bodies connected with pulley system.

Solution:
$$a = \frac{6g-4g}{6+4} = 2\,m/s^2$$
$$T = \frac{2m_1m_2g}{m_1+m_2} = \frac{2\times6\times4\,(10)}{(6+4)} = 48\,N$$

Final Answer: option(2)



43)
$$a = \left(\frac{M_2-M_1}{M_1+M_2}\right)g$$
$$\frac{g}{\sqrt{2}} = \left(\frac{M_2-M_1}{M_1+M_2}\right)g$$
$$(M_1+M_2) = \sqrt{2}M_2 - \sqrt{2}M_1$$
$$\frac{M_1}{M_2} = \left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)$$



Chain block system is in equilibrium so

44) $T = 200 + 100 = 300\,N$

45) $$F = \frac{\Delta P}{\Delta t} = \frac{mv - 0}{0.1}$$
$$= \frac{150 \times 10^{-3} \times 20}{0.1} = 30\,N$$

CHEMISTRY

46)

Geometry with axil a equatorial orbitals.

47)

Co-ordinate bond is directional having direction from donor atom to acceptor atom.

48)

According to dipole moment.

49)

Question Explanation:
We are asked: Which option shown the correct order of boiling points among the given molecular?

Concept:
Boiling point increases with molecular mass and strength of intermolecular forces(van der Waals, polarity, H-bonding).

Solution:
Option 1 : $I_2 > Cl_2 > Br_2 \rightarrow$ incorrect (actual order: $Cl_2 < Br_2 < I_2$).
Option 2 : $He > H_2 \rightarrow$ incorrect (He b.p.≈ 4 K, H_2 b.p ≈ 20 K $\rightarrow H_2 >$ He).
Option 3 : $ICI > Br_2 \rightarrow$ correct (ICI is polar, stronger dipole forces → higher b.p. than non-polar Br_2.)
Option 4 : $H_2 > D_2 < T_2 \rightarrow$ incorrect (heavier isotopes have stronger van der Waals → order is $H_2 < D_2 < T_2$).

Final answer: Option(3)

50)

NH_3 ------ C_6H_6 is an example of dipole induced dipole attraction.

51)

Bent rule

52)

NO_3^- $sp^2\pi$ in $p\pi$-$p\pi$

53)

	N_2^+	N_2
BO	2.5	3.0

54)

Due to πp–πp repulsion bond energy decreases.

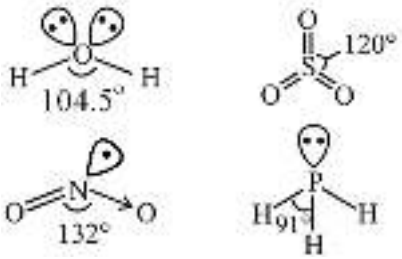
55)

Paramagnetic is due to unpaired 'e' nor because of bond order.

56)

H_2O extent - 4
HF extent - 2

57)



58)

μ CH_3F < CH_3Cl
NH_3 is pyramidal while BF_3 is trigonal planar

59)

NO_2 = 134°
NO_2^- < 120°
NO_3^- = 120

60)

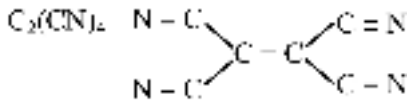
Acc. to MOT-(bond order ∝ stability)

61)

SiF_4 if trihedral while SF_4 is see-saw

62)

Highest boiling point is of SbH_3 and least is of PH_3

63) 

64)

PH_3 → E.N. of P ≅ EN of H

65)

as per MOT configuration.

66) PO_4^{-3} B.O. = 5/4 = 1.25 and no lone pair on P.

67)

XeF_4 shape is square planar

68)

ClF_3 in Bent T shape

69)

wrt H valance in a period shows |23432|

70)

p-nitrophenol BP is more due to inter H-bonding hence less volativle.

71)

$BeCl_2$ solid polymer-due to coordinate bond

72)

Only SiO_2 not show resonance

73)

Concept Based

74)

Boling point ↑ volatile nature ↓

75)

1. Moles of O-atom : $\frac{32g}{16g/mol} = 2mol$
2. Moles of e^- added : 2.
3. Number of electrons : $4 \times N_A$. The statement claims $4N_A$ electrons are added. This is correct.

Statement-II is correct.

Final answer : 2

76)

1. Initial moles : $n_{O_2} = \frac{m}{32}; nN_2 = \frac{m}{28}$
2. Limiting reagent is the reactant with the smallest ratio of $\frac{\text{Initial moles}}{\text{Stoich. Coeff.}}$ (Stoich. Coeff. is 1 for both).
3. Comparison : Since $\frac{m}{32} < \frac{m}{28}, nO_2 < nN_2$

Final Answer : 1

77)

Step-by-Step Calculation: 1. Convert volume from milliliters to liters: 500 mL = 0.500 L. 2. Calculate moles required: $n = 0.240 \text{ mol/L} \times 0.500 \text{ L} = 0.120 \text{ mol}$ 3. Calculate mass of AgNOâ‚ƒ needed: $\text{mass} = 0.120 \text{ mol} \times 170 \text{ g/mol} = 20.4 \text{ g}$

78) **Solution:** Balanced Reaction: $2NH_3(g) \rightarrow 1N_2(g) + 3H_2(g)$
According to stoichiometry (volume ratio): 2 L NH_3 produces 1 L N_2 + 3 L H_2 = 4 L products.
Total final volume (V_{final}) of products from 10 L NH_3; $V_{final} = 10\ L \times \left(\frac{4L\ products}{2L\ reactions}\right) = 20\ L$
Change in volume (ΔV) : $\Delta V = V_{final} - V_{initial} = 20\ L - 10\ L = +10\ L$.
Final answer: option (3)

79) **1. Calculate moles of each element :**
Moles of C = $\frac{3.6}{12} = 0.3$
Moles of H = $\frac{0.7}{1} = 0.7$
Moles of N = $\frac{1.4}{14} = 0.1$
Moles of O = $\frac{1.6}{16} = 0.1$
2. Find the simplest mole ratio (divide by the smallest value, 0.1) :
$C : \frac{0.3}{0.1} = 3$
$H : \frac{0.7}{0.1} = 7$
$N : \frac{0.1}{0.1} = 1$
$O : \frac{0.1}{0.1} = 1$
3. The empirical formula is C_3H_7NO
Final Answer : 1

80) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
1.12 L
Mole of $CO_2 = \frac{1.12}{22.4} = \frac{1}{20}$ mol
Mole of $CaCO_3$ required for 50% yield = $\frac{1}{20} \times \frac{2}{1} = \frac{1}{10}$ mol
Pure $W_{CaCO_3} = \frac{1}{10} \times 100 = 10g$
$W_{sample} = \frac{10}{80} \times 100 = 12.5$ g

81) Let % of oxygen = x
% C = 1.5 × % of O = 1.5 × x
% of H = $\frac{1}{6}$ × % of C = $\frac{1.5x}{6}$

element	%	t. mass (A)	%/A	Simple ratio
C	1.5 x	12	$\frac{1.5x}{12}$	$\frac{1.5x}{12} \times 16 = 2$
H	$\frac{1.5x}{6}$	1	$\frac{1.5x}{6}$	$\frac{1.5x}{6} \times 16 = 4$
O	x	16	$\frac{x}{16}$	= 1

Empirical formula = C_2H_4O

82) I oxide = MO

	M	:	O
mol :	$\frac{77.78}{M}$	:	$\frac{22.22}{16}$ = 1 : 1

⇒ M = 56

II oxide = ?

	M	:	O
mol :	$\frac{70}{56}$	:	$\frac{30}{16}$

= 2 : 3 ⇒ M_2O_3

83) **Calculation**

X(44) = 2 × 220

=> $x = \frac{440}{44}$

= 10

Final Answer 10

Correct option (2)

84) **Solution:** According to Avogadro's law equal volumes of gases at same temperature and pressure contain equal number of molecules irrespective of gas type hence H_2 and SO_2 have equal molecules.

Final Answer: Option (3)

85)

Let the atomic mass of element X be M g/mol.

The molecular mass of $X_4O_6 - 4M + 6(16) - 4M + 96$.

Mass of X in one mole of $X_4O_6 - 4M$.

Given that 10 g of X_4O_6 contains 5.72 g of X :

$$\frac{\text{Mass of X}}{\text{Total mass of compound}} - \frac{4M}{4M+96}$$

$$\frac{5.72}{10} - \frac{4M}{4M+96}$$

$$0.572(4M+96) - 4M$$

$$2.288M + 54.912 - 4M$$

$$1.712M - 54.912$$

$$M - \frac{54.912}{1.712} \approx 32\text{amu}$$

Final Answer: Option (1)

86) $\frac{4.92}{1000} \times 1 = \frac{40 \times 10^{-3}}{M} \times 0.0821 \times 300;$

M $\simeq$ 200; Atomic mass of X = 100.

87)

Solution: Raoult's law is obeyed only when
A–A ≈ B–B ≈ A–B intermolecular attractions.
Final Answer: Option (3) the forces of attraction between like molecules are identical with those between unlike molecules.

88)

1. In the pure state, the mole fraction of water is 1.
2. When acetone is added, the mole faction of water (X_{water}) becomes less than 1 because some of the "space" at the surface and in the bulk is now occupied by acetone molecules.
3. Since $P_{water} = P^{\circ}_{water}$ (a number less than 1), the resulting partial pressure of water must be less than the vapor pressure of pure water.
Final answer: option (3)

89)

For ideal solutions:
• $\Delta H_{mix} = 0$
• $\Delta V_{mix} = 0$
• $\Delta S_{mix} > 0$
• $\Delta G_{mix} < 0$
• They do NOT form azeotropes
• They obey Raoult's law at all compositions

(1) $\Delta S_{mix} = 0 \rightarrow$ Incorrect (Actually $\Delta S_{mix} > 0$)
(2) $\Delta G_{mix} = 0 \rightarrow$ Incorrect (Actually $\Delta G_{mix} < 0$)
(3) They form azeotropes → Incorrect (Ideal solutions do not form azeotropes)
Thus all given statements are incorrect.
Final Answer: Option (4)

90)

Solution: According to Raoult's law, the vapor pressure of a solution is the sum of the vapor pressures of its components. When a solution shows a positive deviation, the intermolecular forces between the different components (A–B) are weaker than those within the pure components (A–A and B–B). This leads to a higher than expected vapor pressure, and since a higher vapor pressure corresponds to a lower boiling point, the mixture boils at a temperature lower than either of its components. This type of mixture is called a minimum boiling azeotrope.
Final Answer: (3)

BIOLOGY

91)

NCERT-XI, Pg # 121

92)

Explain Question:

Longest phase of meiosis?

Concept:

Prophase - I

Solution:

Prophase-I is the longest because pairing, crossing over, and chiasma formation occur.

Final answer:

Option 1

93) **Explanation :**
The question presents a matching exercise related to the stages of mitosis.
Concept :
the stages of mitosis
Solution:
(A) Anaphase - (iii) Chromatid move to poles (B) Telophase - (iv) Nuclear envelope assembles (Reappear) (C) Prophase - (i) Initiation of chromatin condensation(D) Metaphase - (ii) Chromatin condensation complete

Final answer :A-iii, B-iv, C-i, D-ii

94) **correct answer (2)**

95) NCERT (XI) Pg # 168, Para 10.4.1

96) NCERT XIth Pg.#167

97) NCERT (XIth) Pg. # 165, 166

98) NCERT XI Pg. # 163, Fig. 10.1

99) NCERT Pg. # 168

100) NCERT Pg. # 169 Fig 10.3 (c)

101) NCERT Pg. # 126

102) NCERT Pg. # 168

103)

Question Explanation – The question asked to evaluate two statements about Telophase I and Telophase II in meiosis and determine which are correct.

Concept –Based on meiosis.

Solution –

Statement I → Correct (chromosomes partially disperse in Telophase I). Telophase I: Chromosomes do undergo some dispersion, but they do not reach the fully extended state of the interphase nucleus. This is a characteristic feature of Telophase I.

Statement II → Correct (Telophase II leads to tetrad of cells). Telophase II: After this stage, cytokinesis occurs, resulting in the formation of four haploid cells (tetrad of cells).

Final answer – Both statement-I and statement-II are correct

104) **Explain Question:** Question asks about the kingdom showing features: unicellular, aquatic,
autotrophic/heterotrophic, binary fission, eukaryotic.
Concept: This question is based on Protista
Solution: Monera: Unicellular but prokaryotic.
Protista: Unicellular, eukaryotic, mostly aquatic, can be autotrophic (algae) or heterotrophic (protozoa), reproduce asexually.
Fungi: Mostly multicellular (except yeast), heterotrophic only.
Animalia: Multicellular, heterotrophic.
These features fit Protista.
Final answer: 2

105) NCERT Pg # 20,21,22

106) **Explain Question:**
The question asks about the kingdom under which Mycoplasma are classified.

Concept:
Mycoplasma

Solution:
Mycoplasma are prokaryotic organism, lacking a true nucleus and membrane-bound organelles. All prokaryotes are placed under Kingdom Monera (as per NCERT/Whittaker classification)

Final answer: Option (3)

107)

NCERT XI Pg no. 15

108)

Explain Question:
Question asks to find the incorrect match between fungal classes and their examples.
Concept:
This question is based on Fungal classes
Solution :
Incorrect match = Phycomycetes – Albugo, Colletotrichum.
Final answer : (4)

109)

NCERT XI Page No. # 13

110)

Question Explanation: Identifying a fungal class from its features.
Concept: Characteristics of Fungi Classes.
Solution: The key features are the presence of septate and branched mycelium (A), and reproduction mainly by conidia (D). These, along with being saprophytes/parasites (B) and contributing to mineral cycling (C), are the defining characteristics of the class Deuteromycetes (Imperfect Fungi).
Final Answer: Deuteromycetes

111)

Explain Question: Question asks to identify the correct match from the three columns.

Concept: This question is based on Fungal classes

Solution: Albugo → Phycomycetes, obligate parasite → (1-c-i).

Agaricus → Basidiomycetes, edible mushroom → (2-d-iv).

Aspergillus → Ascomycetes, lab weed → (3-a-iii).

Alternaria → Deuteromycetes, decomposer of litter → (4-b-ii).

Final answer: 4

112) **Question Explanation:** Identifying the statement that is factually incorrect about lichens and their environmental sensitivity.

Concept: Lichens as Bioindicators.
Solution: Lichens are a classic **symbiotic association** (mutualism) between a fungus and an alga.

A. **Show mutually useful association between two organisms: Correct**. This is **mutualism** where both partners benefit (alga provides food, fungus provides shelter/water).
B. **Algal component is known as phycobiont: Correct**. The photosynthetic partner is the **phycobiont** (from *phycos* = alga).
C. **They always grow in polluted areas: Incorrect**. This is the opposite of the truth. Lichens are extremely **sensitive to air pollution** (especially SO2) and are used as **bioindicators** because they **fail to grow** or die in polluted environments.
D. **Fungi provides water in lichen: Correct**. The fungal component (mycobiont) absorbs and retains moisture and mineral nutrients for the algal partner.

The statement that they always grow in polluted areas is incorrect.

Final Answer: Option (3)

113)

Explain Question : This question asks to identify the incorrect statement from the four options provided regarding viruses.

Concept : This question is based on virus.

Solution : "Viruses contain either DNA or RNA as their genetic material but never both."

Final Answer : option (2)

114)

Explain Question : Question is asking which infectious agent is a **naked RNA particle** that causes disease in plants.

Concept : This question is based on viroids.

Solution : Plant diseases by naked infectious RNA are caused by viroids.

Final answer : option (1)

115)

NCERT-XI, Pg # 23, 2.3.1

116)

NCERT-XI, Pg # 17-18

117) NCERT XI, Page no. # 20, 21

118) **Explain question:**
The question asks whether the Assertion that Desmids are members of Protista is true, and whether the reason given about their characteristics correctly explains the assertion.

Concept:
Desmids; Protista → Algae; NCERT Biology, chapter: The Unit of Life / algae.

Solution:
Assertion (A): True – Desmids are unicellular, eukaryotic, photosynthetic protists.
Reason (R): False – Desmids are mostly freshwater and non–flagellated, so the reason given (biflagellate, mostly marine) is incorrect.

Final answer: option (3)

119)

Explain Question : The question asks to identify the incorrect statement(s) from a list of four descriptions related to different types of sexual reproduction in algae and other organisms.

Concept : This question is based on Algae, General Characters.

Solution :

"Both Volvox and Fucus show isogamous type of sexual reproduction." - incorrect.

While some species within the genus Volvox can exhibit isogamy, Fucus (a brown alga) is known to exhibit oogamy, not isogamy.

Final Answer : option (4)

120) **Explain Question :** The question asks to identify the commercial product used for growing microbes and in the preparation of ice creams and jellies.
Concept : This question is based on Rhodophyceae
Explanation : “Agar is a polysaccharide obtained from the cell walls of red algae. It is widely used as a solidifying agent in microbiological culture media and in the food industry for preparing jellies and ice-creams.”
Final answer : 1

121)

NCERT-XI, Pg # 32

122)

Explain Question : The question asks in which group of plants the term prothallus is used to describe the gametophyte stage
Concept : This question is based on Pteridophyta, General Characters, Life Cycle And Heterospory
Explanation : "The gametophytes in pteridophytes are inconspicuous, small but multicellular,

free-living, mostly photosynthetic and thalloid structures called prothalli."
Final answer : 1

123)

NCERT-XI, Pg # 30

124)

Explain Question:
Question asking to match column I with column II about classes of pteridophytes and examples.
Concept:
This question is based on Pteridophytes-classification
Solution:
A. Green algae : iv. Ulothrix
B. Lycopsida : iii. Selaginella
C. Horse tail : ii. Equisetum
D. Pteropsida : i. Dryopteris
Final answer: (1)

125)

Explain Question : The question provides four biological statements (a–d) related to plant groups (bryophytes, pteridophytes, and gymnosperms), and asks you to identify the option that includes only the correct statements.

Concept : This question is based on bryophytes, pteridophytes, and gymnosperms

Solution :

(a) Unlike bryophytes and pteridophytes, in gymnosperms the male and female gametophyte do not have an independent free living existence.
This statement is correct. In gymnosperms, the gametophytes are highly reduced and dependent on the sporophyte, unlike the independent gametophyte generation seen in bryophytes and pteridophytes.

(b) In some pteridophytes, the female gametophyte is retained on the parent sporophyte for variable period.
This statement is correct. This retention of the female gametophyte on the sporophyte is a precursor to seed habit, observed in some pteridophytes.

(c) Zygote develops into embryo, retained with in female gametophyte in some pteridophytes.
This statement is correct. The development of the zygote into an embryo within the female gametophyte, which is retained on the parent sporophyte, is a significant evolutionary step towards seed habit in some pteridophytes.

Final Answer : option (4)

126)

Explain Question : The question asks you to identify the single incorrect statement from a list of four descriptions related to the characteristics of gymnosperms.
Concept : This question is based on gymnosperm, general characters and life cycle.
Explanation : All gymnosperms are dioecious without any exception. - Incorrect.
For example, Pinus is monoecious (both male and female cones on the same plant), while Cycas is dioecious.
Final answer : 4

127)

Pinus endosperm is haploid & sieve cell become diploid.

128)

Explain Question : The question asks you to evaluate two separate statements about bryophytes and determine if each is true or false
Concept : Bryophyta, General Characters And Life Cycle
Explanation :
Statement A: Bryophytes live in soil but depend on water for sexual reproduction. - correct.
Bryophytes, like mosses and liverworts, are terrestrial plants found in damp, shady areas. Their sexual reproduction relies on water for the transfer of male gametes (sperm) to the female gametes (egg) .
Statement B: The male gametes (sperm) of bryophytes are biflagellated. - correct.
The male gametes of bryophytes are typically biflagellated, meaning they possess two flagella that enable them to swim through water to reach the archegonium and fertilize the egg
Final answer : 2

129) NCERT (XI) Pg. # 31

130) NCERT (XI) Pg. # 39

131)

Explain Question : The question asks you to identify the single correct statement among the four options provided.

Concept : This question is based on key features of pteridophytes (ferns) and bryophytes (mosses and liverworts).

Solution :
Statement : Protonema is a green, creeping, branched and filamentous stage in mosses, which develops directly from a spore.
This statement is correct. The protonema is a juvenile, filamentous stage in the life cycle of mosses that develops directly from a germinating spore

Final Answer : option (3)

132)

NCERT-XI, Pg # 32

133)

NCERT-XI, Pg # 30

134) **Explain Question :**The question asks for the correct description of the flagella on the zoospores of brown algae.
Concept : This question is based on Phaeophyceae
Explanation : The gametes are pyriform (pear-shaped) and bear two laterally attached flagella of unequal length.
Final answer :4

135) NCERT Page No. 38

136)

NCERT-XI, Pg # 30-33

137) NCERT Pg. # 30

138) NCERT-XI, Pg. # 35, 36

139)

Explain Question : The question asks to identify the correct statements describing the characteristics of gymnosperms from a list of five options (a, b, c, d, e).
Concept : This question is based on characteristics of gymnosperms.
Explanation :
(a) In Cycas, male cone and megasporophyll are borne on different trees. - Correct
Cycas is dioecious, so male and female cones occur on different trees.
(b) Development of pollen grain takes place in microsporophyll. - Correct
Microsporophylls bear microsporangia, where pollen grains (microspores) develop.
(c) Gymnosperms produce haploid microspore and megaspores. - Correct
Gymnosperms undergo meiosis to produce haploid microspores and haploid megaspores.
(e) Pollen grain is released from the microsporangium. - Correct
Pollen grains develop and are released from the microsporangia.
Final answer : 1

140)

Explain Question : The question asks us to identify the incorrect statements about ribosomes from the given list
Concept : This question based on Ribosomes
Explanation : (i) Smallest membrane bound organelle → Incorrect. Ribosomes are the smallest non-membranous organelles.
(ii) They are composed of RNA and proteins → Correct.
(iii) In eukaryotes 70S ribosomes are present in mitochondria → Correct (and also in chloroplasts).
(iv) In prokaryotes, 70S ribosomes are associated with plasma membrane → Correct (loosely attached in cytoplasm and to membrane in some cases, especially during protein synthesis).
(v) All the components of ribosome are synthesised in nucleolus → Incorrect. In nucleolus, rRNA is synthesised and initial ribosome assembly occurs, but ribosomal proteins are synthesised in the cytoplasm.
Final answer : (2) (i) and (v)

141)

Question Explanation : Identify which cell organelles are part of the endomembrane system.

Concept : Endomembrane system

Solution : The correct answer is option 3

A. Endoplasmic Reticulum (i) → Part of endomembrane system

B. Vacuoles (ii) → Part of endomembrane system

C. Mitochondria (iii) → Not part (independent double membrane organelle)

D. Plastids (iv) → Not part (independent double membrane organelle)

E. Peroxisome (v) → Not part (independent organelle)

F. Golgi body (vi) → Part of endomembrane system

G. Lysosome (vii) → Part of endomembrane system

Thus, the correct set is: i, ii, vi, vii
Final Answer : Option 3

142) NCERT (XI) Pg. # 136, Para–5

143) **Explain Question:**
The question asks which prokaryote lacks a cell wall around the cell membrane.

Concept:
This question is based on Mycoplasma
Explanation:
Mycoplasma → A group of prokaryotes (bacteria-like) that lack a cell wall. Their plasma membrane has sterols for protection instead.
Final Answer: (4)

144) NCERT-XI, Pg. # 95

145)

Explain question: This is a "Match the Column" type question about cell organelles and their functions.

Concept: Endoplasmic Reticulum, Golgi apparatus, Cytoskeleton

Solution: .

(a)	**Rough endoplasmic reticulum (RER)**	(iii)	Studded with ribosomes, involved in **protein synthesis**.
(b)	**Smooth endoplasmic reticulum**	(i)	Lacks ribosomes, involved in **lipid synthesis**.
(c)	**Golgi apparatus**	(ii)	Functions in **modification, packaging, and secretion**.
(d)	**Cytoskeleton**	(iv)	Network of microtubules and microfilaments, maintains **cell shape**.

Final answer: option (2)
a-iii, b-i, c-ii, d-iv

146) NCERT (XI) Pg. # 132, para-4

147)

Explain Question : The question asks which organelle-function pair is correct for the given diagram (likely showing a cellular organelle).
Concept : This question is based on Golgi complex
Explanation : Golgi complex – Modifies proteins, forms glycoproteins, sorts and packages them.
Final answer : 3

148)

NCERT Pg. # 93, 94, 95, 96

149) NCERT Pg.#137 & 138 (8.5.9)

150)

Explain Question : four chromosome types (a, b, c, d) and asked to match each with its centromere position class: metacentric, submetacentric, acrocentric, or telocentric.
Concept : This question is based on chromosome
Explanation :
A. Acrocentric: centromere close to one end; one very short arm.
B. Submetacentric: centromere slightly off-center; arms unequal (L-shaped).
C. Metacentric: centromere in the middle; arms equal (V-shaped in anaphase).
D. Telocentric: centromere at the terminal end; effectively one visible arm (I-shaped).
Final answer : 2

151) NCERT-XI, Pg. # 16

152) NCERT-XI, Pg. # 15

153) NCERT-XI, Pg. # 15

154) NCERT-XI, Pg. # 28

155) NCERT-XI, Pg. # 26

156) NCERT-XI, Pg. # 15

157) NCERT-XI, Pg. # 33

158) NCERT-XI, Pg. # 16

159) NCERT-XI, Pg. # 31

160) NCERT-XI, Pg. # 17

161) NCERT-XI, Pg. # 33

162) NCERT-XI, Pg. # 29

163) NCERT-XI, Pg. # 28

164) NCERT-XI, Pg. # 15

165) NCERT-XI, Pg. # 28-31

166) NCERT-XI, Pg. # 28-31

167) NCERT-XI, Pg. # 32

168) NCERT-XI, Pg. # 28-32

169) NCERT-XI, Pg. # 32

170) NCERT-XI, Pg. # 29

171) NCERT-XI, Pg. # 13-15

172) NCERT-XI, Pg. # 33

173) NCERT-XI, Pg. # 30-32

174) NCERT-XI, Pg. # 27

175) NCERT-XI, Pg. # 29

176) NCERT-XI, Pg. # 30

177) NCERT-XI, Pg. # 33

178) NCERT-XI, Pg. # 19

179) NCERT-XI, Pg. # 19

180) NCERT-XI, Pg. # 15